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https://doi.org/10.1063/5.0300977

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Cite as: Phys. Fluids **37**, 113601 (2025); doi: 10.1063/5.0300977
Submitted: 5 September 2025 · Accepted: 16 October 2025 ·
Published Online: 3 November 2025

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ABSTRACT

The simplified thermal lattice Boltzmann method (STLBM), as an emerging approach within the lattice Boltzmann framework, demonstrates exceptional potential for simulating challenging thermal flows at high Reynolds/Rayleigh numbers. While its superior numerical stability in extreme flow conditions has been documented, the influence of lattice velocity models on STLBM's performance remains underexplored in the literature. This study aims to fill this gap through comprehensive investigations into the convergence, accuracy, stability, and computational efficiency of STLBM with different lattice velocity models. Quantitative analyses reveal that while the model's accuracy is only marginally affected, its numerical stability exhibits a significant model dependency. The von Neumann stability analysis is conducted to provide theoretical explanations for the varied performance of numerical stability at low relaxation parameters with different lattice velocity models, in which multi-directional disturbances are considered and more abundant information is provided. It is also found that, although the reduced number of lattice velocities can expect 14%–16% reduction of computational time in each iteration, the total simulation time could be additionally influenced by the varied convergence rates exhibited by different lattice velocity models. Finally, based on multi-criteria evaluation of stability-efficiency tradeoffs, guidelines for the choice of the thermal lattice velocity model in STLBM simulations of different thermal flows are provided.

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I. INTRODUCTION

Incompressible thermal flows are commonly encountered in engineering applications.[1–3] With the development of modern computer technology, numerical simulation has become an effective tool for such problems. Among various simulation methods, the lattice Boltzmann method (LBM) presents appealing characteristics, such as the kinetic nature, simplicity, and explicitness.[4–8] Based on the initial lattice Boltzmann schemes proposed for isothermal flow conditions, many strategies have been proposed to incorporate the thermal effects, among which the double-distribution-function approach is a widely used one.[9–13] Despite its manifest advantages, the conventional LBM with the Bhatnagar–Gross–Krook (BGK)[14] model is subject to a number of disadvantages. These include, but are not limited to, high memory requirements, lack of numerical stability at high Reynolds/Rayleigh numbers, and a cumbersome implementation of physical boundary conditions.

To overcome these limitations, the simplified lattice Boltzmann method (SLBM) was originally developed by Chen *et al.*[15–17] and was later extended to thermal flow problems[18–20] (the simplified thermal lattice Boltzmann method, STLBM). The distinct characteristic of the SLBM/STLBM is the direct evolution of macroscopic variables within the lattice Boltzmann framework. This fundamental shift in the solution algorithm directly addresses the key challenges of the conventional LBM. First, it circumvents the necessity to store a large set of distribution functions, thereby reducing the memory consumption. Second, it allows the direct implementation of physical boundary conditions, thereby simplifying the modeling process. Third, the predictor–corrector scheme of SLBM/STLBM remains explicit, thereby preserving the simplicity and explicitness of the conventional LBM. Furthermore, the SLBM/STLBM demonstrates much better numerical stability in extreme flow conditions at high Reynolds/Rayleigh numbers.

Due to these merits, the STLBM has been applied to various thermal flow problems. Dai *et al.* constructed an enthalpy-based simplified lattice Boltzmann model[21] and later applied it to analyze phase change processes[22] inside the horizontal shell and tube energy storage unit

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with different T-shaped fins, achieving 97.72% enhancement of melting efficiency while maintaining excellent numerical stability at $Ra = 10^6$. In scenarios of high-intensity convection, Ma *et al.*[23] implemented the STLBM to simulate hybrid nanofluid flow and heat transfer over a wide range of high Rayleigh numbers from 10^6 to 10^9, which revealed the transition of flow pattern from symmetric steady state to asymmetric unsteady state. The accurate and stable simulation in these challenging cases confirms the potential of STLBM in high-fidelity simulation of turbulent heat transfer problems. To further extend STLBM's applicability to complex geometries,[24] the immersed boundary-simplified thermal lattice Boltzmann method (IB-STLBM) was later proposed to simulate the flow pattern and energy transfer around a starfish-inspired cylinder. The results are then analyzed to obtain the optimal geometric shape for cooling devices.

Although the above studies demonstrate the capability of STLBM in a broad range of thermal flow problems, one should admit that the selection of lattice velocity model may potentially alter its performance. Despite the fact that the effects of different lattice velocity models on the conventional LBM have been investigated,[25–28] only the D2Q9 and the D3Q15 models have been applied in all kinds of simulations carried out by the STLBM.[18–20] This preference is likely attributable to two primary factors: first, the well-documented concerns regarding the numerical stability of the reduced lattice velocity sets (e.g., D2Q4, D2Q5) in the conventional LBM; and second, the fact that the theoretical stability foundation of STLBM, as established in prior work, was specifically derived for the D2Q9 model, leaving the analysis for alternative models unsupported. This paper will address this concern and provide a thorough investigation into the performance of the STLBM with different lattice velocity models. Numerical tests of benchmark examples will be carried out. Together with the theoretical analysis, comprehensive discussions of the convergency, accuracy, stability and efficiency will be conveyed. Based on these discussions, guidelines for the choice of lattice velocity model will be provided.

The rest part of this paper is organized as follows: Sec. II briefs the basics of STLBM and introduces various lattice velocity models. Section III details the results of the von Neumann stability analysis. Benchmark tests are presented in Sec. IV to thoroughly investigate the performance of STLBM with different lattice velocity models. Final conclusions are drawn in Sec. V.

II. SIMPLIFIED THERMAL LATTICE BOLTZMANN METHOD

The essential idea of the simplified thermal lattice Boltzmann method is to reconstruct solutions to the macroscopic governing equations recovered from the thermal lattice Boltzmann equation with double distribution functions.[29] The formulations of the STLBM reflect direct evolution of the macroscopic variables in a predictor–corrector manner, which gives[19,30]

Predictor:

$$\rho^* = \sum_\alpha f_\alpha^{eq}(\boldsymbol{r} - \boldsymbol{e}_\alpha \delta t,\, t - \delta t), \tag{1}$$

$$(\rho \boldsymbol{u})^* = \sum_\alpha \boldsymbol{e}_\alpha \cdot f_\alpha^{eq}(\boldsymbol{r} - \boldsymbol{e}_\alpha \delta t,\, t - \delta t), \tag{2}$$

$$T^* = \sum_\alpha g_\alpha^{eq}(\boldsymbol{r} - \boldsymbol{e}_\alpha \delta t,\, t - \delta t). \tag{3}$$

Corrector:

$$\rho = \rho^*, \tag{4}$$

$$\rho \boldsymbol{u} = (\rho \boldsymbol{u})^* + (\tau_v - 1)\left[\sum_\alpha \boldsymbol{e}_\alpha \cdot f_\alpha^{eq}(\boldsymbol{r} + \boldsymbol{e}_\alpha \delta t,\, t) - \rho(r, t - \delta t)\boldsymbol{u}(r, t - \delta t)\right] + \boldsymbol{F}_E \delta t, \tag{5}$$

$$T = T^* + (\tau_c - 1)\left[\sum_\alpha \boldsymbol{e}_\alpha \cdot g_\alpha^{eq}(\boldsymbol{r} + \boldsymbol{e}_\alpha \delta t,\, t) - T(r, t - \delta t)\right], \tag{6}$$

where f and g are the density distribution function and the thermal distribution function, respectively; α is the index of the lattice velocity; $\boldsymbol{r}$ and t denote the space location and time level, respectively; ρ, $\boldsymbol{u}$, and T represent the density, the velocity vector, and the temperature, respectively; the superscript "eq" denotes the equilibrium distribution function; δt is the streaming time step; and $\boldsymbol{F}_E$ represents the buoyancy force. According to the Boussinesq approximation, the buoyancy force term is evaluated as follows:

$$\boldsymbol{F}_E = \begin{bmatrix} 0 \\ -\rho g \beta (T - T_0) \end{bmatrix}, \tag{7}$$

where g is the gravitational acceleration; β is the thermal expansion coefficient; and T_0 denotes the average temperature. τ_v and τ_c denote the single relaxation parameters, which are related to the kinematic viscosity ν and the thermal diffusivity χ, respectively, by

$$\nu = c_s^2(\tau_v - 0.5)\frac{\delta x^2}{\delta t}, \tag{8}$$

$$\chi = c_s^2(\tau_c - 0.5)\frac{\delta x^2}{\delta t}, \tag{9}$$

where δx and δt are, respectively, the streaming distance and the time step. Note that, in this paper, the passive scalar transport model[10,31] is considered, which means that the compression work and the viscous heating are neglected. Such neglection is acceptable in the present consideration of incompressible flows at low Mach number.

In terms of the selection of lattice velocity models, the D2Q9 model [see Fig. 1(c)] is employed to ensure mass and momentum conservation (i.e., $\rho = \sum_\alpha f_\alpha^{eq} = \sum_\alpha f_\alpha$ and $\rho \boldsymbol{u} = \sum_\alpha \boldsymbol{e}_\alpha f_\alpha^{eq} = \sum_\alpha \boldsymbol{e}_\alpha f_\alpha$) during the simulation of hydrodynamic field. By using the unit streaming distance $\delta x = 1$ and the unit streaming time step $\delta t = 1$, the D2Q9 model gives the following form of equilibrium distribution function and lattice velocity vectors:

$$f_\alpha^{eq}(\boldsymbol{r}, t) = w_\alpha \rho\left(1 + \frac{\boldsymbol{e}_\alpha \cdot \boldsymbol{u}}{c_s^2} + \frac{(\boldsymbol{e}_\alpha \cdot \boldsymbol{u})^2}{2c_s^4} - \frac{\boldsymbol{u}^2}{2c_s^2}\right), \tag{10}$$

$$\boldsymbol{e}_\alpha = \begin{cases} (0,\, 0), & \alpha = 0, \\ (\pm 1,\, 0),\ (0,\, \pm 1), & \alpha = 1,\, 2,\, 3,\, 4, \\ (\pm 1,\, \pm 1), & \alpha = 5,\, 6,\, 7,\, 8, \end{cases} \quad \text{for } D2Q9, \tag{11}$$

where w_α and c_s denote the weighting coefficient and the sound speed, respectively,

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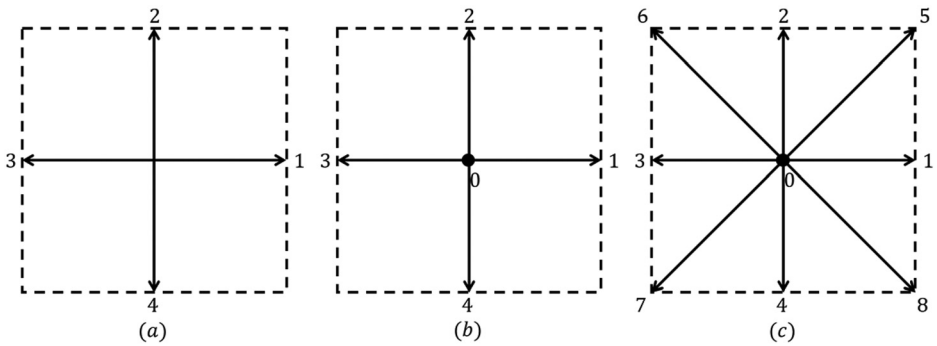


FIG. 1. Illustration of various D2Qq models. (a) D2Q4 model; (b) D2Q5 model; (c) D2Q9 model.

$$w_\alpha = \begin{cases} \frac{1}{9}, & \alpha = 0, \\ \frac{4}{9}, & \alpha = 1, 2, 3, 4, \\ \frac{1}{36}, & \alpha = 5, 6, 7, 8, \end{cases} \quad \text{for } D2Q9, \tag{12a}$$

$$c_s = \frac{1}{\sqrt{3}}. \tag{12b}$$

As for the evolution of the temperature field, multiple lattice velocity models are available for the STLBM in the double-distribution-function (DDF) framework.[29] Apart from the D2Q9 models, the D2Q4 model [Fig. 1(a)] and the D2Q5 model [Fig. 1(b)] are two possible options in two-dimensional simulations. The lattice velocity vectors of these models are given as

$$\boldsymbol{e}_\alpha = (\pm 1, 0), (0, \pm 1), \quad \alpha = 1, 2, 3, 4 \quad \text{for } D2Q4, \tag{13a}$$

$$\boldsymbol{e}_\alpha = \begin{cases} (0, 0), & \alpha = 0, \\ (\pm 1, 0), (0, \pm 1), & \alpha = 1, 2, 3, 4 \end{cases} \quad \text{for } D2Q5. \tag{13b}$$

The equilibrium function g_i^{eq} can either be in a similar form as Eq. (10)

$$g_\alpha^{eq}(\boldsymbol{r}, t) = w_\alpha T\left(1 + \frac{\boldsymbol{e}_\alpha \cdot \boldsymbol{u}}{c_s^2} + \frac{(\boldsymbol{e}_\alpha \cdot \boldsymbol{u})^2}{2c_s^4} - \frac{\boldsymbol{u}^2}{2c_s^2}\right), \tag{14a}$$

or as the following linear form:[32]

$$g_\alpha^{eq}(\boldsymbol{r}, t) = w_\alpha T\left(1 + \frac{\boldsymbol{e}_\alpha \cdot \boldsymbol{u}}{c_s^2}\right). \tag{14b}$$

In the above formulations, w_α and c_s denote the weighting coefficient and the sound speed, respectively,

$$w_\alpha = 1/4, \quad \alpha = 1, 2, 3, 4 \quad \text{for } D2Q4, \tag{15a}$$

$$w_\alpha = \begin{cases} \frac{1}{3}, & \alpha = 0, \\ \frac{1}{6}, & \alpha = 1, 2, 3, 4, \end{cases} \quad \text{for } D2Q5, \tag{15b}$$

$$c_s = \frac{1}{\sqrt{3}}. \tag{15c}$$

In this paper, Eq. (14b) is used for the consideration of simplicity and efficiency.

The STLBM is notable for its direct evolution of macroscopic variables, as well as its excellent stability at low relaxation parameters. Although preliminary work has been carried out by Chen *et al.*[29] with regard to the D2Q9 model, the versatility of the STLBM with different thermal lattice velocity models remain unexplained. Thorough discussions will be made in Sec. III to answer this question.

III. COMPREHENSIVE STABILITY ANALYSIS OF STLBM

A. Isothermal flow condition

The von Neumann stability analysis is a useful strategy to reveal the stability performance of the numerical scheme, which was introduced to the analysis of LBM by Sterling *et al.*[33] It has been demonstrated that the standard LBM could be highly unstable when the relaxation parameter τ is closed to 0.5. Before moving on to the investigation of the thermal models, let us first revisit the isothermal case based on the D2Q9 model. To begin with, let us denote

$$y_1 = \rho, \quad y_2 = \rho u, \quad y_3 = \rho v, \tag{16a}$$

$$y_1^* = \rho^*, \quad y_2^* = (\rho u)^*, \quad y_3^* = (\rho v)^*, \tag{16b}$$

and the vector of variables can be written as

$$\boldsymbol{Y} = (y_1, y_2, y_3)^T, \tag{17a}$$

$$\boldsymbol{Y}^* = \left(y_1^*, y_2^*, y_3^*\right)^T. \tag{17b}$$

With the derivations presented in Ref. 16, one can obtain the following differential equations:

$$\overline{d\boldsymbol{Y}^{n+1}} = \boldsymbol{G} \cdot \overline{d\boldsymbol{Y}^n}, \tag{18}$$

where n and $n+1$ denote the current and the next time step, respectively; $\boldsymbol{G}_{3\times 3}$ is the characteristic matrix whose maximum eigenvalue quantifies the response of the numerical scheme to the numerical disturbances and thus sheds light on its numerical stability. Generally, the maximum eigenvalue $|\lambda_{max}|$ of the characteristic matrix is preferably to be smaller than 1. Physically, $|\lambda_{max}|$ represents the worst-case amplification factor of a numerical disturbance over one time step. The condition $|\lambda_{max}| \leq 1$ is thus the criterion of the linear stability analysis, ensuring that such errors do not grow unboundedly. Here, λ_{max} can be considered as a function of ρ, u, v, θ_x, θ_y and τ_v, i.e.,

$$\lambda_{max} = f\left(\rho, u, v, \theta_x, \theta_y, \tau_v\right), \tag{19}$$

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where $\theta_x = k_x \delta t$ and $\theta_y = k_y \delta t$. k_x and k_y are the wavenumbers in the x and the y directions, respectively.

Given the extreme difficulty in analytical resolution of the maximum eigenvalue λ_{max}, some rational simplifications can be made to facilitate analysis. Following the suggestion of Sterling *et al.*,[33] the most unstable condition is assumed as that both the velocity vector and the wavenumber vector are in alignment with the horizontal direction. Furthermore, the relaxation parameter is set to the value of 0.5, which corresponds to the extreme condition of an infinite Reynolds number. Subsequently, the maximum eigenvalue can be calculated at varied mean flow velocities $|\boldsymbol{u}|$, while the wavenumber is varied from $-\pi$ to $+\pi$. Figure 2 illustrates the maximum eigenvalues of the SLBM with D2Q9 lattice velocity model at different wavenumbers. It can be seen that all eigenvalues are smaller or equal to 1, which indicates the good numerical stability of SLBM under extreme isothermal flow condition.

It should be admitted that the unidirectional simplification made in the previous analysis can only provide limited information. Given the spatial anisotropy of the distribution of the eigenvalues,[34] it is logical to consider the mean flow velocity and wavenumber vector in different directions to obtain more abundant information about the eigenvalues. Following this idea, we release more freedom to the wavenumber vector by allowing both x and y components to vary between $-\pi$ and $+\pi$. Meanwhile, the direction of the mean flow velocity is still aligned with the wavenumber vector, which can maximize the propagation of the numerical disturbances. With these settings, the maximum eigenvalue magnitude at a given mean flow velocity can now be viewed as a function of the two components of the wavenumber vector, i.e., $\lambda_{max} = f(k_x, k_y)$, which is presented in Fig. 3 as two-dimensional gray-scale images, similar to Ref. 35, with darker colors corresponding to smaller values. Here, to facilitate visualization, the maxima and minima of the gray-scale image are set as 0.4 and 1.0, respectively. In Fig. 3, all eigenvalues are smaller than one within the incompressible limit ($Ma = |\boldsymbol{u}|/c_s < 0.3$). The minimum values are concentrated in the corner areas, which suggests that the

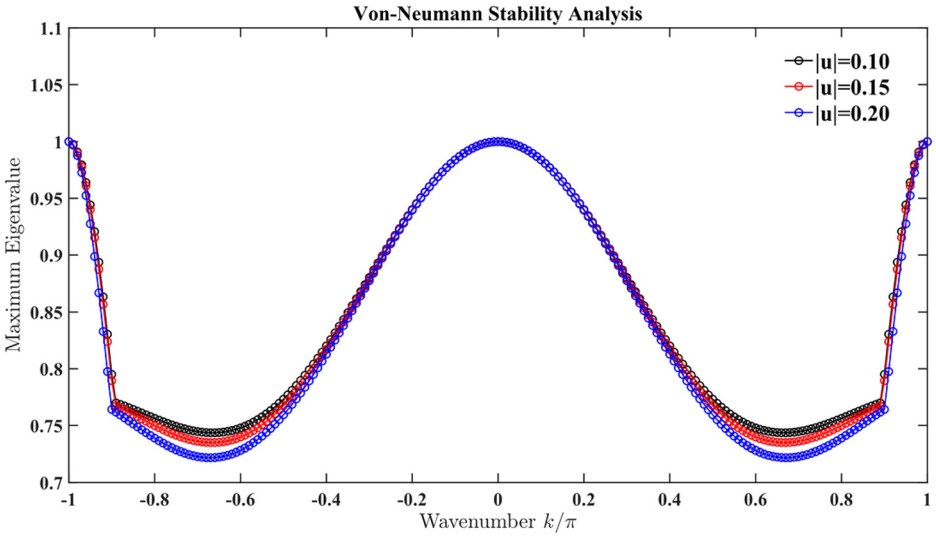
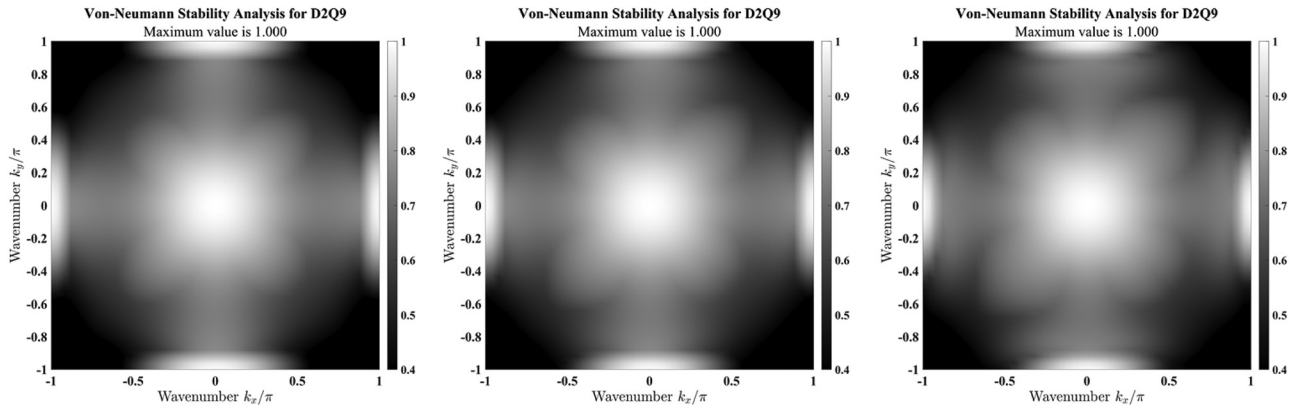


FIG. 2. Maximum eigenvalue magnitudes of the isothermal SLBM as a function of wavenumber k at $\tau_v = 0.5$.



(a) $|u| = 0.10$ (b) $|u| = 0.15$ (c) $|u| = 0.20$

FIG. 3. Maximum eigenvalue magnitude of the SLBM as a function of wavenumber vector under a different mean flow velocity $|\boldsymbol{u}|$ for $\tau_v = 0.5$.

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SLBM is capable to suppress the high-frequency disturbances, i.e., $\left(|k_x|, |k_y|\right) \rightarrow (\pi, \pi)$, and thus sheds light on the underlying mechanisms of its good numerical stability in isothermal condition.

B. Thermal flow condition

In the case of thermal flows, Chen *et al.*[29] carried out the von Neumann stability analysis of the STLBM with D2Q9 model employed for both the density distribution function and the thermal distribution function and with the unidirectional assumption. In this subsection, more comprehensive analysis will be made to shed light on the stability performance of STLBM with different thermal lattice velocity models. To begin with, let us denote

$$y_1 = \rho, \quad y_2 = \rho \boldsymbol{u}, \quad y_3 = \rho v, \quad y_4 = T, \tag{20a}$$

$$y_1^* = \rho^*, \quad y_2^* = (\rho \boldsymbol{u})^*, \quad y_3^* = (\rho v)^*, \quad y_4^* = T^*, \tag{20b}$$

and the vector of variables can be written as

$$\boldsymbol{Y} = (y_1, y_2, y_3, y_4)^T, \tag{21a}$$

$$\boldsymbol{Y}^* = \left(y_1^*, y_2^*, y_3^*, y_4^*\right)^T. \tag{21b}$$

For the sake of simplicity, the following notations will be employed:

$$\begin{cases} f_\alpha^{eq} = f_\alpha^{eq}(\boldsymbol{r} - \boldsymbol{e}_\alpha \delta t,\ t - \delta t), \\ g_\beta^{eq} = g_\beta^{eq}(\boldsymbol{r} - \boldsymbol{e}_\alpha \delta t,\ t - \delta t), \\ y_{j\alpha/\beta} = y_{j\alpha/\beta}\left(\boldsymbol{r} - \boldsymbol{e}_{\alpha/\beta} \delta t,\ t - \delta t\right), \end{cases} \tag{22}$$

where $\alpha = 0, \ldots, 8$ is the index of the D2Q9 lattice velocity and β is the index of the thermal lattice velocity model.

Substituting Eqs. (20a) and (22) into Eqs. (10) and (14b) yields

$$f_\alpha^{eq} = w_\alpha^v y_{1\alpha} + \frac{w_\alpha^v}{c_s^2}\left(e_{x\alpha} y_{2\alpha} + e_{y\alpha} y_{3\alpha}\right) + \frac{w_\alpha^v}{2c_s^4}\frac{\left(e_{x\alpha} y_{2\alpha} + e_{y\alpha} y_{3\alpha}\right)^2}{y_{1\alpha}} - \frac{w_\alpha^v}{2c_s^2}\frac{y_{2\alpha}^2 + y_{3\alpha}^2}{y_{1\alpha}}, \tag{23a}$$

$$g_\beta^{eq} = w_\beta^c y_{1\beta} + \frac{w_\beta^c}{c_s^2} y_{4\beta}\frac{\left(e_{x\beta} y_{2\beta} + e_{y\beta} y_{3\beta}\right)}{y_{1\beta}}. \tag{23b}$$

Equations 23(a) and 23(b) can be substituted into Eqs. (1)–(3) to obtain a set of nonlinear algebraic equations, which are denoted as

$$\boldsymbol{Y}^* = \boldsymbol{F}\left(y_1^n, y_2^n, y_3^n, y_4^n\right) = \boldsymbol{F}(\boldsymbol{Y}^n), \tag{24}$$

where n represents the time level. In order to apply the von Neumann stability analysis, it is necessary to linearize the above equations. By differentiating Eq. (24) with respect to the variable y_l, we can gain a set of linear equations as follows:

$$\overline{d\boldsymbol{Y}^*} = \boldsymbol{E} \cdot \overline{d\boldsymbol{Y}^n}, \tag{25a}$$

which can also be written as the following index notation form:

$$\overline{dy_j^*} = E_{jl} \cdot \overline{dy_l^n},\ j, l = 1, 2, 3, 4, \tag{25b}$$

where

$$(E_{jl})_{4\times 4} = \left(\frac{\partial y_j^*}{\partial y_l}\right)^n = \begin{bmatrix} \partial_{11}^* & \partial_{12}^* & \partial_{13}^* & 0 \\ \partial_{21}^* & \partial_{22}^* & \partial_{23}^* & 0 \\ \partial_{31}^* & \partial_{32}^* & \partial_{33}^* & 0 \\ \partial_{41}^* & \partial_{42}^* & \partial_{43}^* & \partial_{44}^* \end{bmatrix} \tag{26}$$

is the matrix of partial derivatives; $\overline{dy_j^*}$ and $\overline{dy_l^n}$ are the magnitudes of dy_j^* and dy_l^n, respectively. Similar procedures can be applied to the formulations in the corrector step, resulting in the following formulation:

$$\overline{d\boldsymbol{Y}^{n+1}} = \boldsymbol{G} \cdot \overline{d\boldsymbol{Y}^n}, \tag{27a}$$

or in an index notation form as

$$\overline{dy_j^{n+1}} = G_{jl} \cdot \overline{dy_l^n},\quad j, l = 1, 2, 3, 4, \tag{27b}$$

where $\boldsymbol{G}_{4\times 4}$ is the characteristic matrix for the stability analysis. The detailed derivation and the full formulations of the matrices $\boldsymbol{E}$ and $\boldsymbol{G}$ can be found in the Appendix.

Similar to the above analysis for isothermal condition, the maximum eigenvalue λ_{max} of the characteristic matrix derived in the thermal flow condition is essentially a function of ρ, u, v, T, θ_x, θ_y, τ_v, and τ_c, i.e.,

$$\lambda_{max} = f\left(\rho, u, v, T, \theta_x, \theta_y, \tau_v, \tau_c\right), \tag{28}$$

where $\theta_x = k_x \delta t$ and $\theta_y = k_y \delta t$. k_x and k_y are the wavenumber in the x and the y directions, respectively.

Preliminary analysis is first carried out by setting the directions of the velocity and the wavenumber vectors horizontal. The relaxation parameters are set to the extreme value of 0.5 corresponding to infinite Reynolds/Rayleigh numbers. Subsequently, the maximum eigenvalue is calculated at varied mean flow velocities $|\boldsymbol{u}|$, while the wavenumber is swept over from $-\pi$ to $+\pi$.

Figure 4 illustrates the results of the preliminary stability analysis of the STLBM with different thermal lattice velocity models, which provides some interesting insights. The maximum eigenvalues of all three thermal lattice velocity models are smaller than or equal to 1 under the incompressible limit. Such feature suggests that all three models would exhibit good stability in extreme thermal flow conditions. In the meantime, the three thermal lattice velocity models share the common characteristic that the instability is most likely to occur at $k = \pm\pi$, i.e., high-frequency components. Finally, all three thermal lattice velocity models reveal nearly identical maximum eigenvalues at the same mean flow velocity, indicating that numerical stability is almost irrelevant to the thermal lattice velocity model employed. This finding is somewhat perplexing, given the fact that the expression of the characteristic matrix $\boldsymbol{G}$ is primarily determined by the thermal lattice velocity model (see the Appendix). Fortunately, by considering varied directions of the wavenumber vector, we are able to resolve this puzzle and explain the varied stability exhibited by different thermal lattice velocity models.

In Fig. 5, the distributions of the maximum eigenvalues with different thermal lattice velocity models are presented in gray-scale images with darker colors corresponding to smaller values. Compared to the unidirectional wavenumber assumption-based results in Fig. 4, more information about the numerical stability of STLBM is conveyed. The new finding is that, although the eigenvalues remain smaller than 1 (thereby offering further evidence for its numerical stability), different

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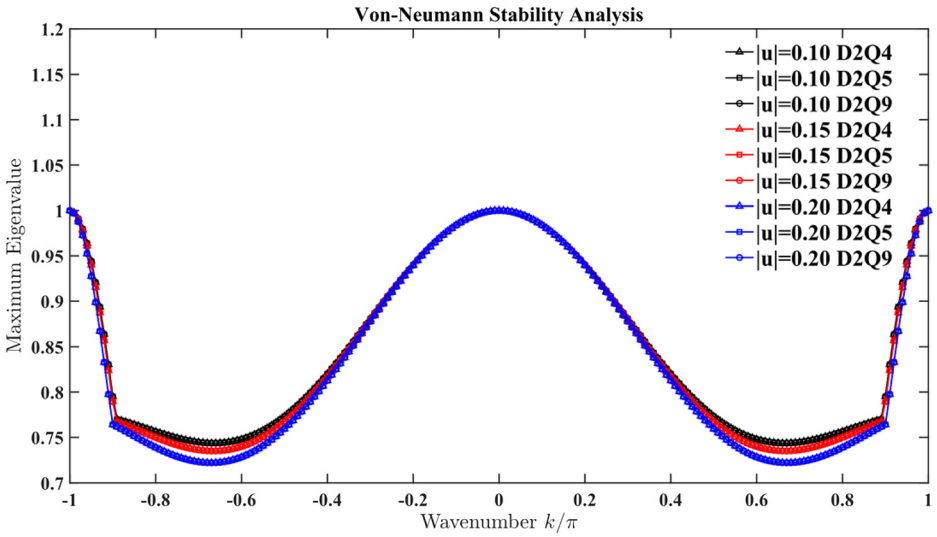


FIG. 4. Maximum eigenvalue magnitudes of the STLBM with different thermal lattice velocity models as a function of wavenumber k at $\tau_v = \tau_c = 0.5$.

lattice velocity models can exhibit distinct eigenvalue distribution patterns. For low-frequency components corresponding to the central region ($\boldsymbol{k} \to 0$), all three models exhibit similar characteristic: the eigenvalues are close to the value of 1. However, for high-frequency components corresponding to the corner regions $\left[\left(|k_x|, |k_y|\right) \to (\pi, \pi)\right]$, the three thermal lattice velocity models present diversified features. For the D2Q4 model, the numerical instability can be a potential concern, as the high-frequency components of disturbance can lead to eigenvalues approaching to the unity. With regard to the D2Q5 and the D2Q9 models, they both demonstrate analogous eigenvalue distribution patterns to those observed in the isothermal case, i.e., the high-frequency components of disturbances are well suppressed. In addition, the numerical stability performance in a wider range of relaxation parameters ($\tau > 1$) will be presented in Sec. IV B 2.

IV. NUMERICAL TESTS

This section thoroughly examinates the convergence, accuracy, numerical stability, and efficiency of the STLBM with different thermal lattice velocity models through two numerical benchmarks, namely, the porous plate problem and the 2D natural convection in a square cavity. The convergence criteria used in all simulations are

$$\begin{cases} \max(V_{err},\, T_{err}) < 10^{-8}, \\ V_{err} = \dfrac{\sqrt{\sum_i \left(\boldsymbol{u}^{t+\delta t}(\boldsymbol{x}_i) - \boldsymbol{u}^{t}(\boldsymbol{x}_i)\right)^2}}{\sqrt{\sum_i \left(\boldsymbol{u}^{t+\delta t}(\boldsymbol{x}_i)\right)^2}}, \\ T_{err} = \dfrac{\sqrt{\sum_i \left(T^{t+\delta t}(\boldsymbol{x}_i) - T^{t}(\boldsymbol{x}_i)\right)^2}}{\sqrt{\sum_i \left(T^{t+\delta t}(\boldsymbol{x}_i)\right)^2}}, \end{cases} \tag{29}$$

where V_{err} and T_{err} are the relative errors of the velocity and the temperature fields, respectively.

A. The porous plate problem

The porous plate problem is a typical benchmark with analytical solution. In this subsection, we use this example to validate the accuracy of all models. The configuration is a fully developed channel flow where the upper cold plate moves at a constant horizontal velocity u_0 and a constant flow at a vertical velocity v_0 is injected through the lower hot plate and withdrawn from the upper plate at the same rate. The size of the computational domain is $L \times H$, where $L = 2H$ and H are the length and the height of the channel, respectively. The left and the right boundaries are periodic, while the velocity and the temperature boundary conditions are applied to the upper and the bottom plates accordingly (see Fig. 6).

The analytical solutions of the horizontal velocity and the temperature distribution are given as follows:

$$u = u_0 + \frac{e^{Re\cdot\frac{y}{H}} - 1}{e^{Re} - 1}, \tag{30a}$$

$$T = T_h - \Delta T \cdot \frac{e^{Pr\cdot Re\cdot y/H} - 1}{e^{Pr\cdot Re} - 1}, \tag{30b}$$

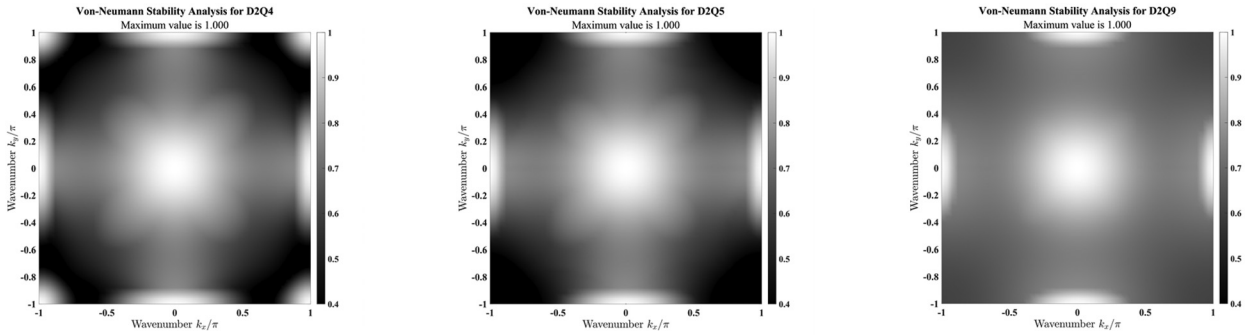
where $Re = v_0 H/\nu$ is the Reynolds number based on the injected velocity v_0 and the channel height H; $Pr = \nu/\chi$ is the Prandtl number; $\Delta T = T_h - T_c$ is the temperature difference between the hot and the cold plates. Another key dimensionless parameter is the Rayleigh number, which is defined as $Ra = g\beta\Delta T H^3/(\nu\chi)$. In our simulations, the following parameters are used:

$$T_h = 1.0, \quad T_c = 0.0, \quad \nu = 0.1, \quad u_0 = 0.1,$$
$$L = 2.0, \quad H = 1.0, \quad Ra = 100.0.$$

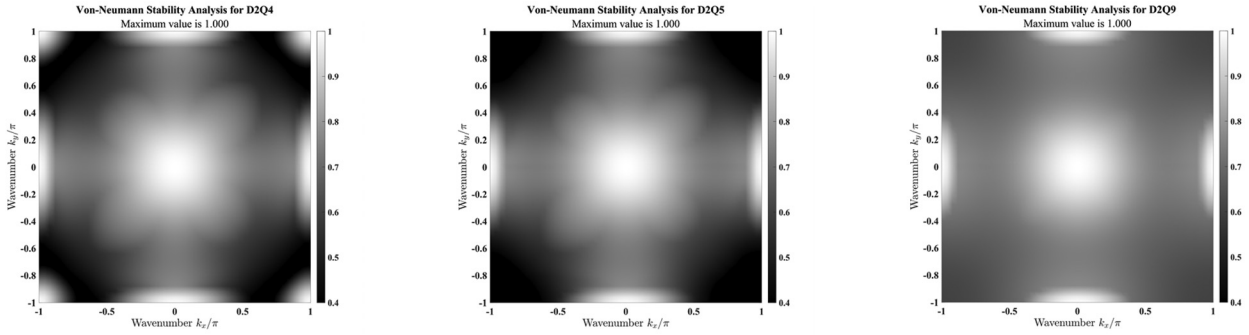
To validate the convergence and the practical order of accuracy, five sets of mesh size ($[41 \times 21]$, $[81 \times 41]$, $[121 \times 61]$, $[161 \times 81]$, and $[201 \times 101]$) are chosen to carry out numerical experiments. The Reynolds number and the Prandtl number are, respectively, set as $Re = 5$ and $Pr = 0.71$, and the injected velocity v_0 is determined by the Reynolds number accordingly. Here, the L_2 norms of the relative

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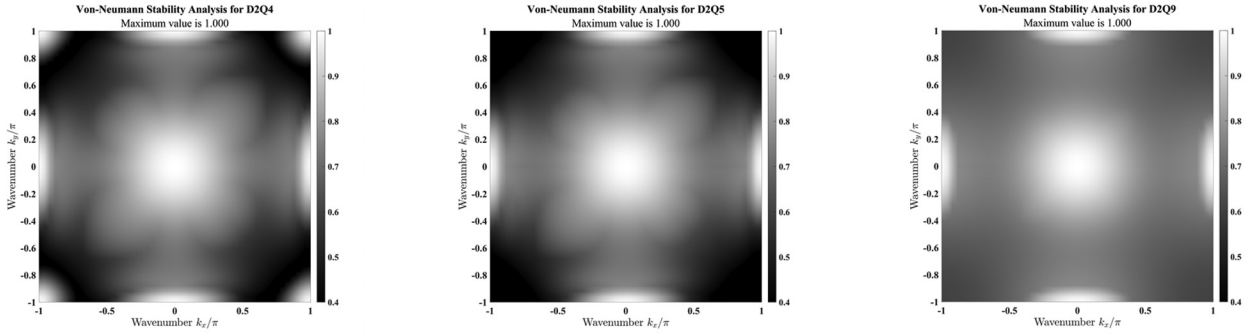
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(a) Stability analysis for D2Q4 (left), D2Q5 (middle) and D2Q9 (right) with the mean flow velocity $|\boldsymbol{u}| = 0.10$



(b) Stability analysis for D2Q4 (left), D2Q5 (middle) and D2Q9 (right) with the mean flow velocity $|\boldsymbol{u}| = 0.15$



(c) Stability analysis for D2Q4 (left), D2Q5 (middle) and D2Q9 (right) with the mean flow velocity $|\boldsymbol{u}| = 0.20$

FIG. 5. Maximum eigenvalue magnitude of the STLBM with different thermal lattice velocity models as a function of wavenumber vector under different mean flow velocity $|\boldsymbol{u}|$ for $\tau_v = \tau_c = 0.5$.

errors of the horizontal velocity u and the temperature T are, respectively, defined as

$$Err_V = \frac{\sqrt{\sum_i \left(u^{numerical}(\boldsymbol{x}_i) - u^{analytical}(\boldsymbol{x}_i)\right)^2}}{\sqrt{\sum_i \left(u^{analytical}(\boldsymbol{x}_i)\right)^2}}, \tag{31a}$$

$$Err_T = \frac{\sqrt{\sum_i \left(T^{numerical}(\boldsymbol{x}_i) - T^{analytical}(\boldsymbol{x}_i)\right)^2}}{\sqrt{\sum_i \left(T^{analytical}(\boldsymbol{x}_i)\right)^2}}, \tag{31b}$$

where the superscripts "numerical" and "analytical" denote the solutions of STLBM and the analytical solutions, respectively. Figure 7 presents the L_2 norms of the relative errors for different thermal lattice

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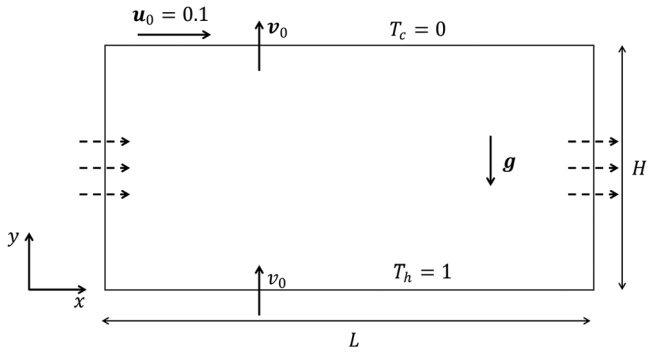


FIG. 6. Configuration of the porous plate problem.

velocity models. It can be seen that the slopes of the fitted lines range from 1.9 to 2.0, which indicates that the STLBM with different thermal lattice velocity models are all approximately second-order accurate. It is also noteworthy that the values of the relative errors for all thermal lattice velocity models are very close, which indicates comparable accuracy.

To further validate the present method, more simulations under various *Re* and *Pr* are carried out with the finest mesh in Fig. 7. Figures 8–11 illustrate the normalized velocity and temperature profiles for $Pr = 0.71$ with $Re = [5, 10, 20]$ and for $Re = 10$ with $Pr = [0.2, 0.8, 1.5]$, respectively. It can be seen that the results are in good agreement with the analytical solutions.

B. Natural convection in a square cavity

The natural convection in a square cavity is simulated in this subsection to provide more thorough evaluation of the models in two-dimensional conditions. The physical configuration is shown in Fig. 12. The square cavity $(L \times L)$ is bounded by the left hot wall with the temperature of T_h and the right cold wall with the temperature of T_c. The top and the bottom walls are assumed to be adiabatic.

Two important dimensionless parameters, namely, the Rayleigh number *Ra* and the Prandtl number *Pr*, determine the flow features, which are, respectively, defined as

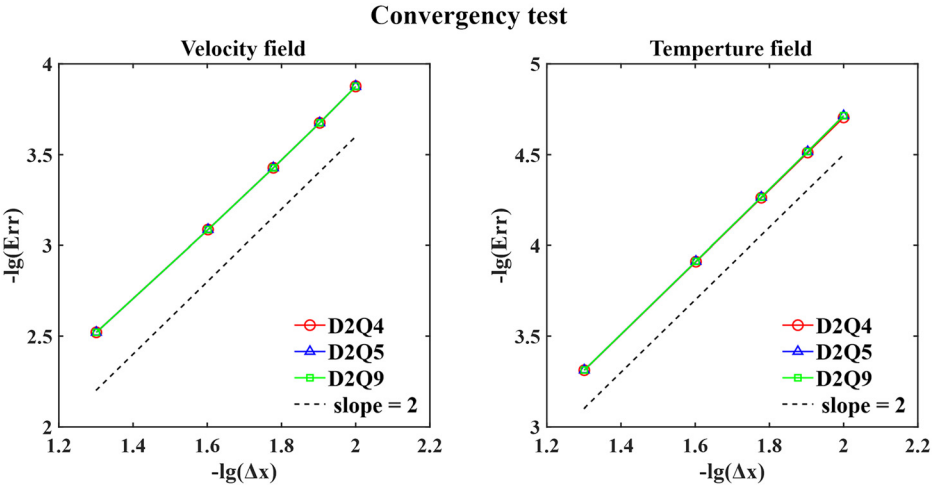


FIG. 7. Relative errors of the velocity(left) and the temperature(right) at Re = 5. Slopes of the fitting lines for [D2Q4, D2Q5, D2Q9] are [1.93, 1.93, 1.93] (left) and [1.99, 2.00, 2.00] (right).

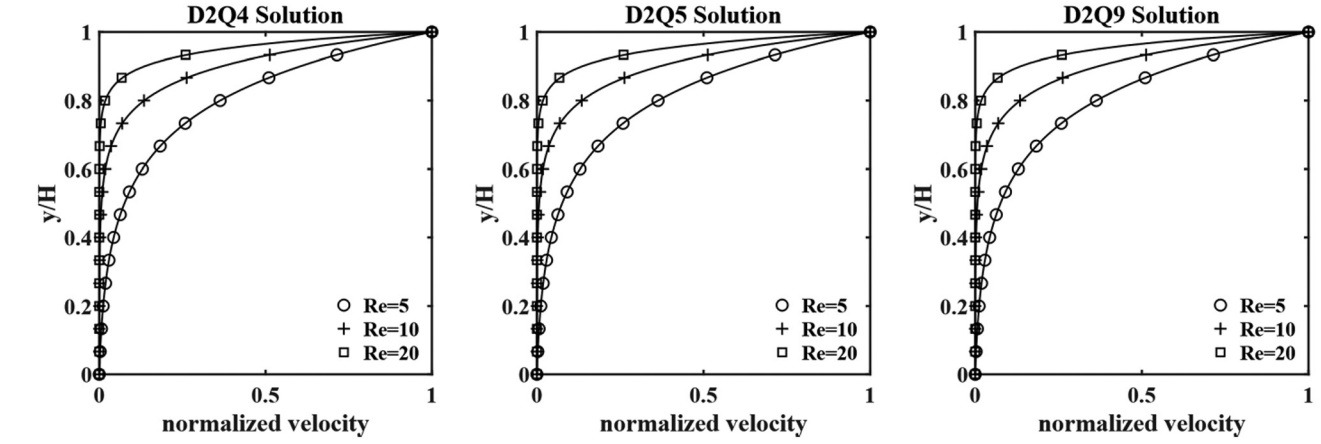


FIG. 8. Distributions of the normalized velocity u/u_0 along the vertical centerline with the D2Q4 model (left), the D2Q5 model (middle), and the D2Q9 model (right) at $Pr = 0.71$, $Re = [5, 10, 20]$. The black solid lines represent the analytical solution.

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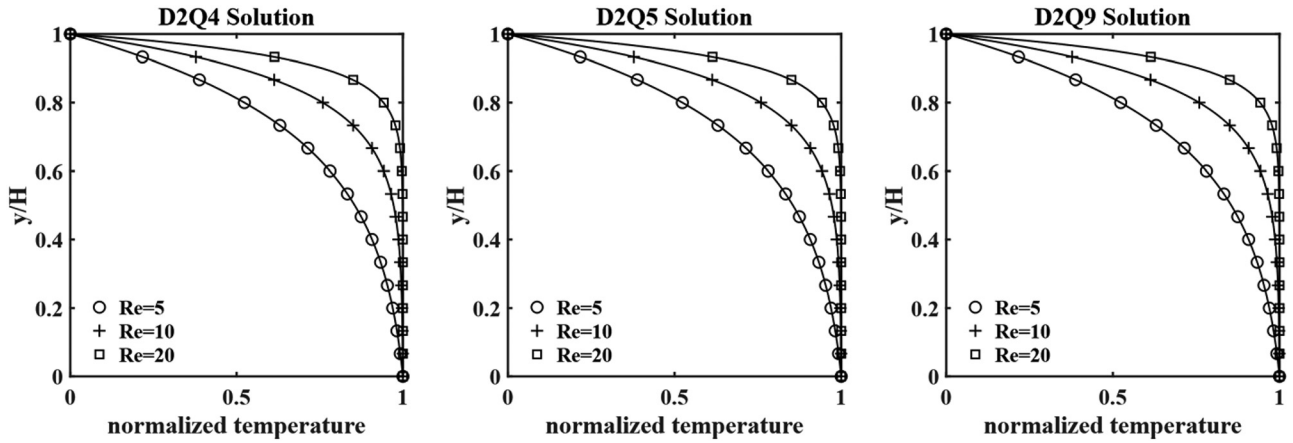


FIG. 9. Distributions of the normalized temperature $(T - T_c)/\Delta T$ along the vertical centerline with the D2Q4 model (left), the D2Q5 model (middle), and the D2Q9 model (right) at $Pr = 0.71$, $Re = [5, 10, 20]$. The black solid lines "-" represent the analytical solution.

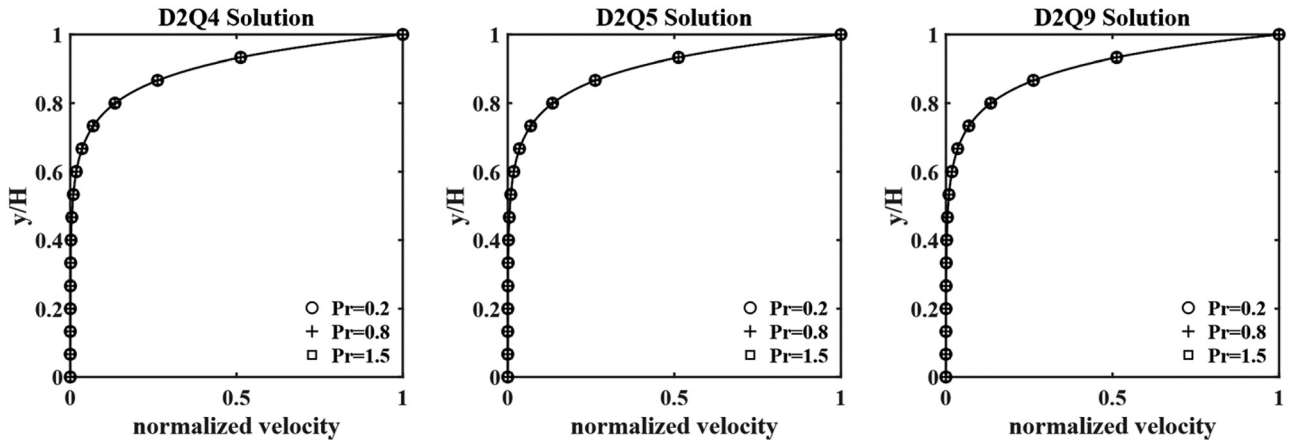


FIG. 10. Distributions of the normalized velocity u/u_0 along the vertical centerline with the D2Q4 model (left), the D2Q5 model (middle) and the D2Q9 model (right) at $Re = 10$, $Pr = [0.2, 0.8, 1.5]$. The black solid lines "-" represent the analytical solution.

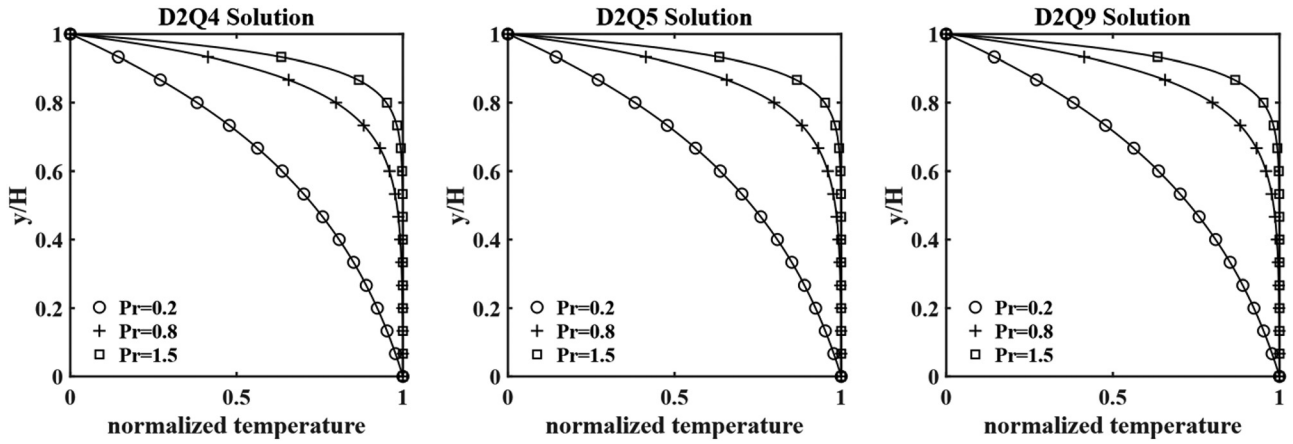


FIG. 11. Distributions of the normalized temperature $(T - T_c)/\Delta T$ along the vertical centerline with the D2Q4 model (left), the D2Q5 model (middle) and the D2Q9 model (right) at $Re = 10$, $Pr = [0.2, 0.8, 1.5]$. The black solid lines "-" represent the analytical solution.

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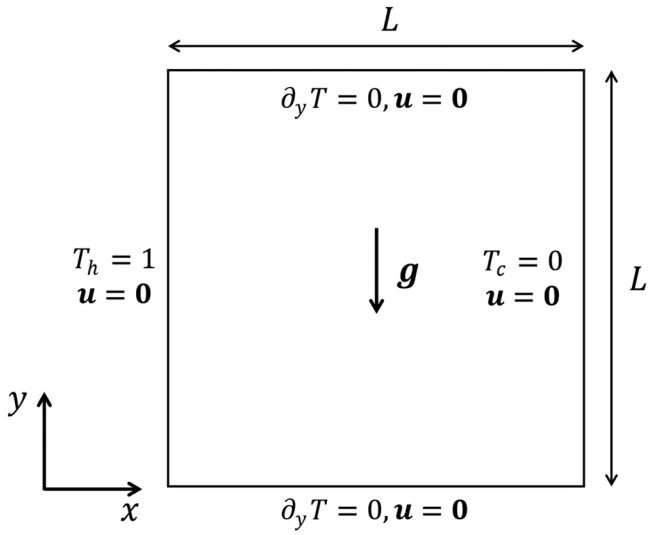


FIG. 12. Configuration of the 2D natural convection problem.

$$Ra = \frac{g\beta \cdot \Delta T \cdot L^3}{\nu \cdot \chi} = \frac{V_c^2 \cdot L^2}{\nu \cdot \chi}, \tag{32}$$

$$Pr = \frac{\nu}{\chi}, \tag{33}$$

where $\Delta T = T_h - T_c$ is the temperature difference between the hot and the cold walls; $V_c = \sqrt{g\beta\Delta TL}$ is the characteristic velocity. In our simulations, Pr and V_c are, respectively, set to be 0.71 and 0.1, while a wide range of Ra is considered. To facilitate quantitative comparisons, the volume average Nusselt number $\overline{Nu}$, the mean Nusselt number $\overline{Nu_0}$ on the hot wall, and the local Nusselt number Nu are defined as follows:

$$\overline{Nu} = \frac{L}{\chi\Delta T} \cdot \frac{1}{L^2} \cdot \int\int \left(uT - \chi \frac{\partial T}{\partial x} \right) d\Omega, \tag{34}$$

$$\overline{Nu_0} = \frac{1}{\Delta T} \int \left. \frac{\partial T}{\partial x} \right|_{x=x_0} dy, \tag{35}$$

$$Nu = \frac{L}{\Delta T} \frac{\partial T}{\partial x}. \tag{36}$$

1. Convergence and accuracy

In our simulations, three typical Rayleigh numbers, namely, $\mathrm{Ra} = 10^4$, 10^5, and 10^6, are considered. Numerical simulations with different mesh sizes are carried out to evaluate the performance of the STLBM with different thermal lattice velocity models. Tables I–III present the computational results of hydrodynamic field at different Rayleigh numbers, in which u_{max} and v_{max} are the maximum horizontal and the maximum vertical components of the dimensionless velocity $\boldsymbol{u}'$ along the vertical and the horizontal centerlines, respectively; y and x are the dimensionless locations of these maximum velocity components. The dimensionless velocity $\boldsymbol{u}'$ is defined as

$$\boldsymbol{u}' = \frac{\boldsymbol{u}L}{\chi}. \tag{37}$$

TABLE I. The STLBM results of the hydrodynamic properties in 2D natural convection problem with various thermal lattice velocity models at $Ra = 10^4$, $Pr = 0.71$. The symbol "-" denotes the numerical divergence.

Mesh size	Lattice model	u_{max}	y	v_{max}	x
11×11	D2Q4	13.996	0.800	14.389	0.200
	D2Q5	13.431	0.800	13.833	0.200
	D2Q9	13.597	0.800	13.960	0.200
51×51	D2Q4	16.120	0.820	19.515	0.120
	D2Q5	16.116	0.820	19.509	0.120
	D2Q9	16.129	0.820	19.526	0.120
101×101	D2Q4	16.166	0.820	19.614	0.120
	D2Q5	16.163	0.820	19.613	0.120
	D2Q9	16.170	0.820	19.618	0.120
201×201	D2Q4	16.181	0.825	19.634	0.120
	D2Q5	16.181	0.825	19.633	0.120
	D2Q9	16.182	0.825	19.635	0.120
301×301	D2Q4	...	...	...	...
	D2Q5	16.186	0.823	19.639	0.120
	D2Q9	16.187	0.823	19.640	0.120
Reference	**DQ**[36]	**16.190**	**0.825**	**19.638**	**0.120**
Relative error	D2Q4	...	...	...	...
	D2Q5	−0.02%	−0.24%	−0.01%	0.00%
	D2Q9	−0.02%	−0.24%	0.01%	0.00%

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TABLE II. The STLBM results of the hydrodynamic properties in 2D natural convection problem with various thermal lattice velocity models at $Ra = 10^5$, $Pr = 0.71$. The symbol "-" denotes the numerical divergence.

Mesh size	Lattice model	u_{max}	y	v_{max}	x
11 × 11	D2Q4	41.451	0.800	43.898	0.200
	D2Q5	36.939	0.800	39.342	0.200
	D2Q9	37.407	0.800	39.729	0.200
51 × 51	D2Q4	33.957	0.860	65.550	0.080
	D2Q5	33.830	0.860	65.241	0.080
	D2Q9	34.165	0.860	65.507	0.080
101 × 101	D2Q4	34.570	0.850	68.127	0.070
	D2Q5	34.561	0.850	68.094	0.070
	D2Q9	34.651	0.850	68.177	0.070
201 × 201	D2Q4	34.733	0.855	68.558	0.065
	D2Q5	34.725	0.855	68.553	0.065
	D2Q9	34.749	0.855	68.573	0.065
301 × 301	D2Q4	34.754	0.853	68.614	0.067
	D2Q5	34.746	0.853	68.611	0.067
	D2Q9	34.757	0.853	68.620	0.067
Reference	**DQ**[36]	**34.736**	**0.855**	**68.640**	**0.065**
Relative error	D2Q4	0.05%	−0.23%	−0.04%	3.07%
	D2Q5	0.03%	−0.23%	−0.04%	3.07%
	D2Q9	0.06%	−0.23%	−0.03%	3.07%

TABLE III. The STLBM results of the hydrodynamic properties in 2D natural convection problem with various thermal lattice velocity models at $Ra = 10^6$, $Pr = 0.71$. The symbol "-" denotes the numerical divergence.

Mesh size	Lattice model	u_{max}	y	v_{max}	x
11 × 11	D2Q4	118.41	0.800	127.93	0.200
	D2Q5	98.002	0.800	107.39	0.200
	D2Q9	98.858	0.800	108.12	0.200
51 × 51	D2Q4	57.106	0.840	184.74	0.060
	D2Q5	56.561	0.840	178.83	0.060
	D2Q9	57.801	0.840	179.84	0.060
101 × 101	D2Q4	61.806	0.850	209.27	0.040
	D2Q5	61.583	0.850	207.83	0.040
	D2Q9	62.445	0.850	208.31	0.040
201 × 201	D2Q4	64.164	0.850	218.85	0.040
	D2Q5	64.130	0.850	218.65	0.040
	D2Q9	64.390	0.850	218.83	0.040
301 × 301	D2Q4	64.618	0.850	219.83	0.037
	D2Q5	64.584	0.850	219.77	0.037
	D2Q9	64.704	0.850	219.85	0.037
Reference	**DQ**[36]	**64.775**	**0.850**	**220.64**	**0.035**
Relative error	D2Q4	−0.24%	0.00%	−0.37%	5.71%
	D2Q5	−0.29%	0.00%	−0.39%	5.71%
	D2Q9	−0.11%	0.00%	−0.36%	5.71%

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In addition, to evaluate the simulation accuracy of thermal behaviors, the volume average Nusselt number $\overline{Nu}$, the mean Nusselt number $\overline{Nu_0}$ on the hot wall, and the maximum and the minimum local Nusselt numbers on the hot wall (Nu_{max} and Nu_{min}) with their vertical coordinates are summarized in Tables IV–VI.

We can see that, if the numerical stability is assured, the STLBM with different thermal lattice velocity models is able to converge to accurate solutions by refining the mesh. As demonstrated in Tables I–III, the relative errors of the converged velocity (or the results obtained on the finest mesh) to the reference data are within 0.4% for all models. In Tables IV–VI, all Nusselt numbers are within the range of solutions obtained by the differential quadrature method[36] (DQ), the thermal lattice Boltzmann method[13] (TLBM), the finite difference method[37] (FDM), and the coupled simplified lattice Boltzmann method[30] (CSLBM). These results demonstrate the marginal effect of the thermal lattice velocity model on the convergence and the accuracy of STLBM simulation of thermal flows.

More evidence of the comparable accuracy of different thermal lattice velocity models is illustrated in Figs. 13 and 14, where the streamlines and the temperature contours obtained by the finest mesh are plotted. As can be seen, most lines are overlapped, which indicates the velocity and the temperature fields given by all models are very similar. The only noticeable discrepancy appears in the upper region near the top wall at $Ra = 10^6$, where the results given by the D2Q4 model and the D2Q5 model slightly deviate from that given by the D2Q9 model.

2. Numerical stability

Although the numerical stability of the STLBM with different thermal lattice velocity models has been demonstrated in Sec. III, the practical performance should be verified in numerical tests. Here, we consider the Rayleigh numbers of $Ra = 10^7$, 10^8, 10^9 and carry out simulations on the very coarse mesh size of 11×11. In this circumstance, the values of the relaxation parameters are very close to 0.5, which is likely to trigger numerical instability. Within the lattice Boltzmann framework, such extreme parametric condition is analogous to the extreme flow condition of very high Reynolds/Rayleigh numbers. As shown in Table VII, our simulations with different models can all ensure convergence, which numerically demonstrate the good numerical stability of STLBM. In the meantime, the comparable performance in numerical stability of different thermal lattice velocity models not only showcases the robustness of the STLBM, but also makes it meaningful to further compare the computational efficiency in Subsection IV B 3.

Apart from the low relaxation parameter scenario, cases with high relaxation parameters could also encounter stability issues.[38] As showcased in Tables I and IV, the STLBM with the D2Q4 model could diverge in the case of $Ra = 10^4$ with the fine mesh size of 301×301, suggesting that different thermal lattice velocity models may exhibit different numerical stability at high relaxation parameters. A series of numerical tests are then designed to answer this question. The acquisition of the high relaxation parameter is realized by using very fine meshes. As demonstrated in Table VIII, the D2Q4, the D2Q5, and the D2Q9 models diverge in sequential order as the relaxation parameters increase.

TABLE IV. The STLBM results of the thermal parameters in 2D natural convection problem with various thermal lattice velocity models at $Ra = 10^4$, $Pr = 0.71$. The symbol "-" is used to denote numerical results that are either divergent or absent from the reference.

Mesh	Lattice model	$\overline{Nu}$	$\overline{Nu_0}$	Nu_{max}	y	Nu_{min}	y
11×11	D2Q4	1.912	1.679	2.722	0.000	0.550	1.000
	D2Q5	1.871	1.729	2.782	0.000	0.572	1.000
	D2Q9	1.881	1.690	2.633	0.000	0.552	1.000
51×51	D2Q4	2.236	2.242	3.575	0.140	0.606	1.000
	D2Q5	2.235	2.242	3.575	0.140	0.607	0.960
	D2Q9	2.236	2.238	3.502	0.140	0.596	1.000
101×101	D2Q4	2.244	2.245	3.549	0.140	0.588	0.980
	D2Q5	2.244	2.245	3.549	0.140	0.588	0.980
	D2Q9	2.244	2.244	3.527	0.140	0.585	0.980
201×201	D2Q4	2.244	2.245	3.536	0.145	0.571	1.000
	D2Q5	2.244	2.245	3.536	0.145	0.571	1.000
	D2Q9	2.244	2.245	3.531	0.145	0.571	1.000
301×301	D2Q4	...	...	...	...	...	...
	D2Q5	2.244	2.245	3.533	0.140	0.563	1.000
	D2Q9	2.244	2.245	3.530	0.143	0.564	1.000
Reference	**TLBM**[13]	...	**2.235**	**3.507**	**0.148**	**0.584**	**0.984**
	FDM[37]	**2.234**	**2.242**	**3.545**	**0.149**	**0.592**	**1.000**
	DQ[36]	**2.245**	**2.248**	**3.543**	**0.145**	**0.586**	**1.000**
	CSLBM[30]	...	**2.251**	**3.542**	**0.145**	...	...

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TABLE V. The STLBM results of the thermal parameters in 2D natural convection problem with various thermal lattice velocity models at $Ra = 10^5$, $Pr = 0.71$. The symbol "-" is used to denote numerical results that are either divergent or absent from the reference.

Mesh	Lattice model	$\overline{Nu}$	$\overline{Nu_0}$	Nu_{max}	y	Nu_{min}	y
11 × 11	D2Q4	4.238	2.245	3.663	0.000	0.675	1.000
	D2Q5	3.797	2.347	3.787	0.000	0.714	1.000
	D2Q9	3.871	2.293	3.568	0.000	0.705	1.000
51 × 51	D2Q4	4.395	4.331	7.401	0.100	0.758	1.000
	D2Q5	4.369	4.345	7.445	0.100	0.770	1.000
	D2Q9	4.375	4.332	7.148	0.100	0.745	1.000
101 × 101	D2Q4	4.498	4.499	7.767	0.080	0.764	0.980
	D2Q5	4.495	4.499	7.768	0.080	0.767	0.980
	D2Q9	4.496	4.495	7.614	0.090	0.753	0.980
201 × 201	D2Q4	4.519	4.519	7.755	0.080	0.741	0.985
	D2Q5	4.518	4.519	7.754	0.080	0.742	0.985
	D2Q9	4.518	4.518	7.705	0.085	0.736	0.990
301 × 301	D2Q4	4.520	4.520	7.738	0.080	0.730	0.990
	D2Q5	4.520	4.520	7.738	0.080	0.730	0.993
	D2Q9	4.520	4.520	7.715	0.083	0.727	0.993
Reference	**TLBM**[13]	...	**4.504**	**7.658**	**0.088**	**0.728**	**0.990**
	FDM[37]	**4.510**	**4.532**	**7.761**	**0.085**	**0.736**	**1.000**
	DQ[36]	**4.523**	**4.527**	**7.788**	**0.080**	**0.725**	**1.000**
	CSLBM[30]	...	**4.564**	**7.787**	**0.080**	...	...

TABLE VI. The STLBM results of the thermal parameters in 2D natural convection problem with various thermal lattice velocity models at $Ra = 10^6$, $Pr = 0.71$. The symbol "-" is used to denote numerical results that are either divergent or absent from the reference.

Mesh	Lattice model	$\overline{Nu}$	$\overline{Nu_0}$	Nu_{max}	y	Nu_{min}	y
11 × 11	D2Q4	10.80	2.482	4.050	0.000	0.760	1.000
	D2Q5	8.931	2.616	4.241	0.000	0.811	1.000
	D2Q9	9.180	2.558	4.003	0.000	0.814	1.000
51 × 51	D2Q4	8.335	6.874	12.530	0.060	0.885	1.000
	D2Q5	8.007	7.018	12.761	0.080	0.900	1.000
	D2Q9	8.019	6.991	12.289	0.080	0.921	1.000
101 × 101	D2Q4	8.565	8.299	15.990	0.050	1.029	1.000
	D2Q5	8.494	8.346	16.138	0.050	1.051	1.000
	D2Q9	8.496	8.330	15.602	0.050	0.991	1.000
201 × 201	D2Q4	8.766	8.750	17.444	0.040	1.047	0.990
	D2Q5	8.757	8.751	17.459	0.040	1.053	0.990
	D2Q9	8.757	8.746	17.126	0.045	1.018	0.990
301 × 301	D2Q4	8.807	8.802	17.590	0.040	1.026	0.990
	D2Q5	8.801	8.802	17.588	0.040	1.027	0.990
	D2Q9	8.801	8.799	17.397	0.040	1.008	0.990
Reference	**TLBM**[13]	...	**8.767**	**17.288**	**0.0441**	**0.998**	**0.990**
	FDM[37]	**8.798**	**8.928**	**18.076**	**0.0456**	**1.005**	**1.000**
	DQ[36]	**8.762**	**8.721**	**16.070**	**0.040**	**1.019**	**1.000**
	CSLBM[30]	...	**8.995**	**17.851**	**0.038**	...	...

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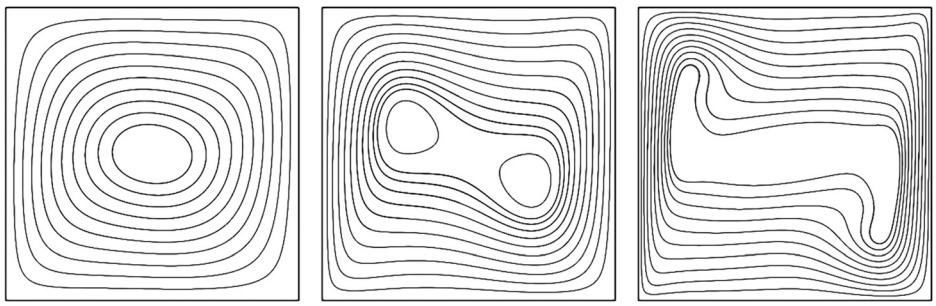


FIG. 13. Streamlines of the 2D natural convection in a square cavity at $Ra = 10^4$ (left), $Ra = 10^5$ (middle), and $Ra = 10^6$ (right). Green lines represent the D2Q4 model, red lines represent the D2Q5 model, and black lines represent the D2Q9 model. Here, most lines are overlapped.

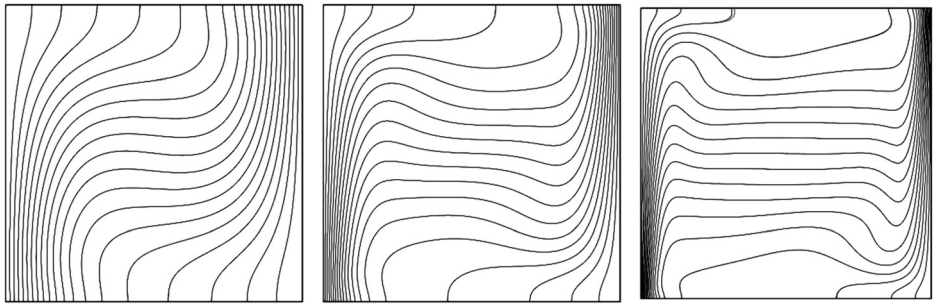


FIG. 14. Temperature contours of the 2D natural convection in a square cavity at $Ra = 10^4$ (left), $Ra = 10^5$ (middle), and $Ra = 10^6$ (right). Green lines represent the D2Q4 model, red lines represent the D2Q5 model, and black lines represent the D2Q9 model. Here, most lines are overlapped.

Next, we tentatively use the von Neumann stability analysis to explain the varied stability performance of the STLBM with different thermal lattice velocity models at high relaxation parameters. The characteristic matrix derived in Sec. III is still the basis of our analysis, while the relaxation parameters in the elements are set to the values listed in Table VIII. The results with the unidirectional assumption for the wavenumber are presented in Fig. 15, which reveals no significant difference and thus contradict to our numerical tests.

To resolve this puzzle, we remove the unidirectional assumption for the wavenumber and consider both two components of the wavenumber vector as variables. The distributions of the maximum eigenvalues are then plotted in the 2D gray-scale images in Fig. 16, in which the darker color corresponds to the smaller value. Furthermore, regions where the maximum eigenvalues exceeding 1 are denoted as the "unstable regions." A common tendency shared by all thermal lattice velocity models is that the numerical instability, which is represented by the area of the unstable regions and the magnitude of the eigenvalues, would grow with the increasing relaxation parameters. However, the distributions of the unstable regions and the maximum eigenvalue exhibit strong dependency on the thermal lattice velocity model employed. As shown in Fig. 16(a), for cases simulated on the mesh size of 301×301, the D2Q4 model presents the largest unstable region at the corner area and a maximum eigenvalue of 1.456. In contrast, the D2Q5 and the D2Q9 models remain stable, which explains the convergence of both models in the numerical test. As the mesh size is refined to 361×361, the relaxation parameter increases, the D2Q5 model generates a new unstable region in the corner areas and the middle area near the edges [see Fig. 16(b)]. The distribution of the

TABLE VII. Stability performance of the STLBM with different thermal lattice velocity models at low relaxation parameters.

Ra	τ_v	τ_c	D2Qq model	Status
10^7	0.500799	0.501126	D2Q4	Converge
			D2Q5	Converge
			D2Q9	Converge
10^8	0.500253	0.500356	D2Q4	Converge
			D2Q5	Converge
			D2Q9	Converge
10^9	0.500080	0.500113	D2Q4	Converge
			D2Q5	Converge
			D2Q9	Converge

TABLE VIII. Stability performance of the STLBM at $Ra = 10^4$, $Pr = 0.71$ with different thermal lattice velocity models at high relaxation parameters.

Mesh	τ_v	τ_c	D2Qq Model	Status
301×301	1.258353	1.568103	D2Q4	Diverge
			D2Q5	Converge
			D2Q9	Converge
361×361	1.410024	1.781724	D2Q4	Diverge
			D2Q5	Diverge
			D2Q9	Converge
401×401	1.511138	1.924138	D2Q4	Diverge
			D2Q5	Diverge
			D2Q9	Diverge

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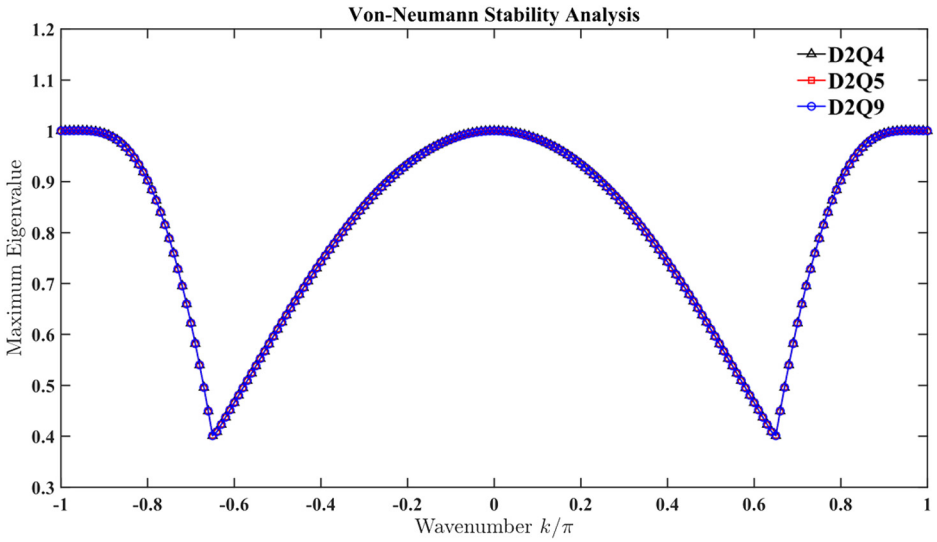


FIG. 15. Maximum eigenvalue magnitude as a function of wavenumber k at certain relaxation parameter correspond to 301 $\times$ 301 mesh and $Ra = 10^4$.

unstable regions exhibited by the D2Q9 model, however, is only located in the middle area near the edges, and its maximum value is lower than that of the D2Q5 model. This observation suggests that the D2Q9 model would be more stable than the D2Q5 model in extreme conditions with high relaxation parameters, which explained the varied stability performance shown in Table VIII. Furthermore, the results from the numerical experiments (Table VIII) and the theoretical analysis (Fig. 16) collectively substantiate the conclusion in Sec. III that the suppression of the high-frequency component plays a pivotal role in enhancing numerical stability, which can be accomplished by increasing the number of lattice velocities.

3. Computational efficiency

This subsection will address the remaining question upon the computational efficiency of the STLBM. By adopting the D2Q4 or the D2Q5 model instead of the D2Q9 model, the number of lattice velocity directions is reduced and less computational time for each iteration can be anticipated.

To showcase the effect of thermal lattice velocity model on the computational efficiency, the natural convection in a square cavity at $Ra = 10^6$ is simulated by the STLBM on the mesh size of 301 $\times$ 301. As anticipated, the computational time required by the D2Q4 model and the D2Q5 model is notably reduced in comparison to that of D2Q9. As shown in Table IX, if the total time consumed by the STLBM simulation with the D2Q9 model is specified as 1.00, simulations with the D2Q4 and the D2Q5 models can approximately reduce 35% and 15% of computational time, respectively. In order to facilitate more comprehensive analysis, the total iteration steps and the computational time consumed by 1000 iteration steps are also provided. The reduction percentage for the D2Q4 model and the D2Q5 model compared to the D2Q9 model in each iteration are 16% and 15%, respectively. It is evident that, apart from the expected reduced computational time in each iteration due to the reduced number of lattice velocity directions, the varied convergence speed exhibited by different thermal lattice velocity models contribute to the computational efficiency. The D2Q4 model presents the most appealing convergence performance in this typical example of thermal flows, which strengthens its advantage in computational efficiency. It is worth noting that the computational efficiency usually depends on the programing strategies, particularly in the context of parallel computation. In this paper, these uncertain factors are excluded, and all simulations are executed in sequential codes.

Based on the comprehensive evaluations of the numerical stability and computational efficiency of the STLBM with different thermal lattice velocity models, we tentatively give the guidelines for the selection of thermal lattice velocity models in different situations, as illustrated in Fig. 17. For thermal flow problems at high Reynolds/Rayleigh numbers and expecting steady flow state, D2Q4 is the optimal choice due to its superior convergency performance. For thermal flows at low Reynolds/Rayleigh numbers or in unsteady state, if the relaxation parameters are not excessively large, the D2Q5 model could be a preferred choice. This is attributed to the comparable computational time per iteration in comparison to the D2Q4 model, as well as the enhanced numerical stability in a broader range of relaxation parameters.

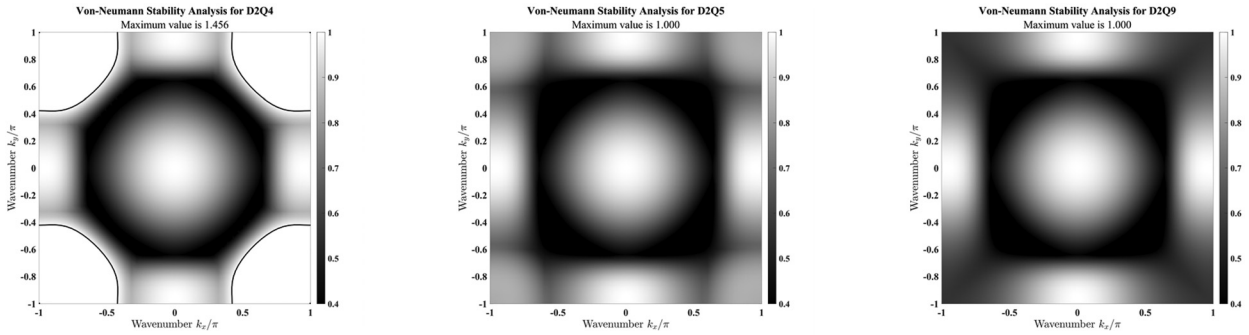
V. CONCLUSIONS

This paper presents a thorough investigation into the performance of the STLBM with different thermal lattice velocity models, with attention paid on the accuracy, numerical stability and computational efficiency.

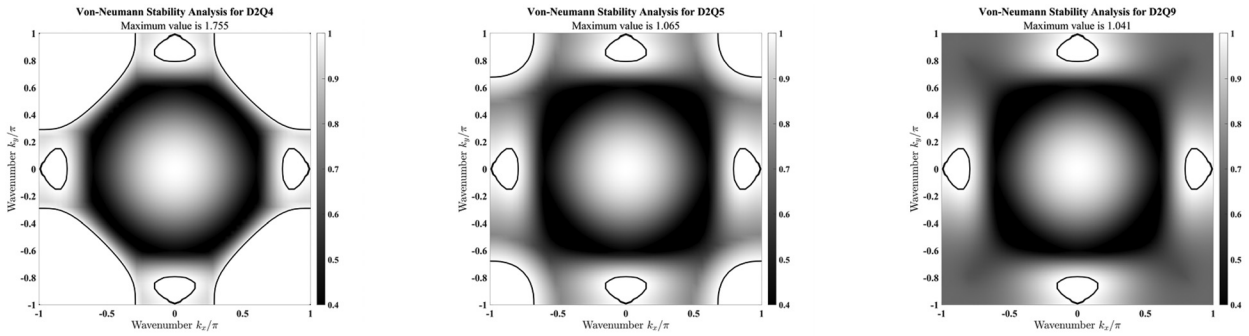
A more comprehensive von Neumann stability analysis is conducted in both isothermal and thermal conditions, which removes the unidirectional assumption made in the previous work and thus provides more abundant information. For the isothermal case, the high-frequency components of disturbance are found to be suppressed by the STLBM, which sheds light on the mechanism contributing to its superior stability. For thermal cases, three thermal lattice velocity models, namely, the D2Q4, the D2Q5, and the D2Q9 models, are considered. With the present stability analysis, the dependency of the numerical stability of STLBM on the thermal lattice velocity models is well revealed. It is found that, in extreme conditions of low or high relaxation parameters, the D2Q9 and the D2Q5 models perform better

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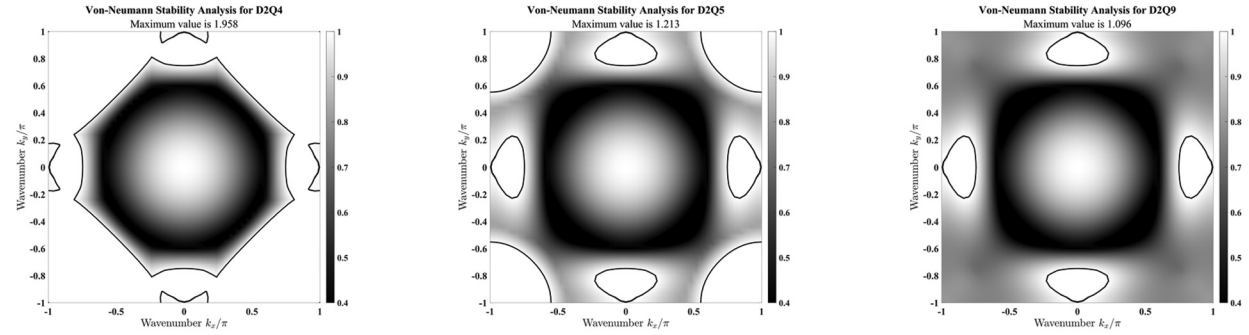
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(a) Stability analysis of the STLBM with the D2Q4 (left), the D2Q5 (middle) and the D2Q9 (right) thermal lattice velocity models on the mesh size of 301×301.



(b) Stability analysis of the STLBM with the D2Q4 (left), the D2Q5 (middle) and the D2Q9 (right) thermal lattice velocity models on the mesh size of 361×361.



(c) Stability analysis of the STLBM with the D2Q4 (left), the D2Q5 (middle) and the D2Q9 (right) thermal lattice velocity models on the mesh size of401×401.

FIG. 16. Maximum eigenvalue magnitude as a function of wavenumber vector $\boldsymbol{k} = (k_x, k_y)$ with different mesh size. The black solid lines "-" mark the "unstable regions" where the maximum eigenvalue magnitude exceeds 1.

than the D2Q4 model in suppressing high-frequency components of disturbance, which favors the numerical stability of STLBM.

Validations and verifications of the accuracy, stability, and efficiency of the STLBM with different thermal lattice velocity models are then carried out by two well-recognized benchmark examples. The second order of accuracy is validated through the simulations of the porous plate problem, and the change of thermal lattice velocity models exhibits marginal influence on the numerical accuracy. As for the example of two-dimensional natural convection in a square cavity, test cases of the STLBM with different thermal lattice velocity models demonstrate satisfying and comparable numerical accuracy. In addition, the theoretical findings in the stability analysis are well supported by the series of tests of this example, i.e., the usage of either the D2Q5 or the D2Q9 model would improve the numerical stability of STLBM in a

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TABLE IX. Computational time of the 2D natural convection problem for different D2Qq model at $Ra = 10^6$ with 301×301 mesh.

Lattice model	Total time	Iterative step	Time per $1000\delta t$
D2Q4	0.65	285 000	0.84
D2Q5	0.85	366 000	0.85
D2Q9	1.00	366 000	1.00

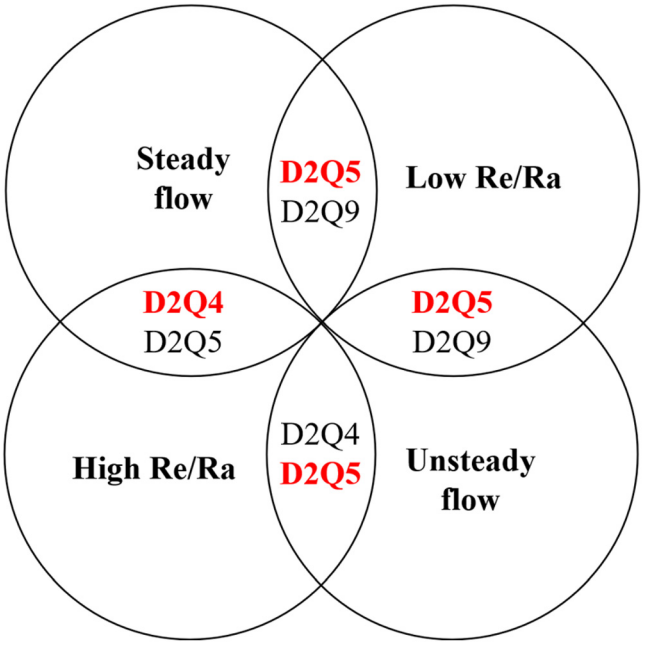


FIG. 17. Recommendations for the selection of various D2Qq models in different situations. Red fonts represent primary choice; black fonts represent secondary choice.

wider range of relaxation parameters. Finally, apart from the expected feature that the D2Q4 and the D2Q5 models can reduce the computational time per iteration due to the reduced number of thermal lattice velocity vectors, the D2Q4 model exhibits the best convergence rate in the present tests and 35% reduction in total computational time.

Despite the evident consistency between the findings of the linear stability analysis and numerical experimentation, which lend support to the conclusions and guidelines drawn in this study, certain limitations should be acknowledged. The theoretical framework employed here is a linear von Neumann analysis, which, while powerful, inherently operates under the assumptions of the homogeneous and periodic flow domain. Therefore, several complex scenarios lie beyond its immediate scope. These include: (1) the analysis of the transient growth due to non-normal effects in the buoyancy-coupled operator, even when the spectral radius indicates stability; (2) the complex interactions between specific wall boundary conditions and the reduced lattice models (e.g., D2Q4), which may alter the stability dynamics in confined flows; and (3) the fully nonlinear development of instabilities in strongly anisotropic or sheared thermal flows. A comprehensive exploration of these phenomena, which constitutes a valuable direction for future research, would necessitate the application of nonlinear stability theories and high-fidelity simulations of flows in complex geometries.

In conclusion, as a general guideline for the choice of thermal lattice velocity model in applications of the STLBM, the D2Q4 model is more suitable for steady flow problems at high Reynolds/Rayleigh number, while the D2Q5 is a preferred option for problems at low Reynolds/Rayleigh numbers or expecting unsteady flow state.

ACKNOWLEDGMENTS

This work was supported by the National Natural Science Foundation of China (Grant Numbers: 52471333 and 52201329) and the Fundamental Research Funds for the Central Universities.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Zheng-Wei He: Data curation (lead); Investigation (lead); Methodology (equal); Validation (lead); Writing – original draft (lead). **Chang Shu:** Conceptualization (supporting); Investigation (equal); Methodology (equal); Supervision (equal); Writing – review & editing (supporting). **Zhen Chen:** Conceptualization (lead); Funding acquisition (lead); Methodology (equal); Project administration (lead); Supervision (equal); Writing – review & editing (lead).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

APPENDIX: VON NEUMANN STABILITY ANALYSIS OF THE STLBM WITH VARIOUS THERMAL LATTICE VELOCITY MODELS

In this section, we will present the detailed derivation of the von Neumann stability analysis for the STLBM with various thermal lattice velocity models. Same notations in Eqs. (16) and (18) are adopted. The specific formulations for the intermediate density y_1^*, moments $y_{2,3}^*$, and temperature y_4^* in Eq. (20) can be expressed as

$$y_1^* = \sum_\alpha \left[w_\alpha^v y_{1\alpha}^n + \frac{w_\alpha^v}{c_s^2}\left(e_{x\alpha} y_{2\alpha}^n + e_{y\alpha} y_{3\alpha}^n\right) + \frac{w_\alpha^v}{2c_s^4}\frac{\left(e_{x\alpha} y_{2\alpha}^n + e_{y\alpha} y_{3\alpha}^n\right)^2}{y_{1\alpha}^n} - \frac{w_\alpha^v}{2c_s^2}\frac{\left(y_{2\alpha}^n\right)^2 + \left(y_{3\alpha}^n\right)^2}{y_{1\alpha}^n}\right], \tag{A1}$$

$$y_2^* = \sum_\alpha \left[w_\alpha^v e_{x\alpha} y_{1\alpha}^n + \frac{w_\alpha^v e_{x\alpha}}{c_s^2}\left(e_{x\alpha} y_{2\alpha}^n + e_{y\alpha} y_{3\alpha}^n\right) + \frac{w_\alpha^v e_{x\alpha}}{2c_s^4}\frac{\left(e_{x\alpha} y_{2\alpha}^n + e_{y\alpha} y_{3\alpha}^n\right)^2}{y_{1\alpha}^n} - \frac{w_\alpha^v e_{x\alpha}}{2c_s^2}\frac{\left(y_{2\alpha}^n\right)^2 + \left(y_{3\alpha}^n\right)^2}{y_{1\alpha}^n}\right], \tag{A2}$$

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$$y_3^* = \sum_\alpha \left[w_\alpha^v e_{y\alpha} y_{1\alpha}^n + \frac{w_\alpha^v e_{y\alpha}}{c_s^2}\left(e_{x\alpha} y_{2\alpha}^n + e_\alpha y_{3\alpha}^n\right) + \frac{w_\alpha^v e_{y\alpha}}{2c_s^4}\frac{\left(e_{x\alpha} y_{2\alpha}^n + e_{y\alpha} y_{3\alpha}^n\right)^2}{y_{1\alpha}^n} - \frac{w_\alpha^v e_{y\alpha}}{2c_s^2}\frac{(y_{2\alpha}^n)^2 + (y_{3\alpha}^n)^2}{y_{1\alpha}^n}\right], \tag{A3}$$

$$y_4^* = \sum_\beta \left[w_\beta^c y_{4\beta}^n + \frac{w_\beta^c}{c_s^2} y_{4\beta}^n \left(\frac{e_{x\beta} y_{2\beta}^n + e_{y\beta} y_{3\beta}^n}{y_{1\beta}^n}\right)\right], \tag{A4}$$

where $\alpha = 0, \ldots, 8$ is the index of the D2Q9 lattice velocity model and β is the index of the thermal lattice velocity vector.

Consider a random component of the variables

$$\begin{cases} dy_j^* = \overline{dy_j^*} \exp(\boldsymbol{ik}\cdot\boldsymbol{r}), \\ dy_j^n = \overline{dy_j^n} \exp(\boldsymbol{ik}\cdot\boldsymbol{r}), \\ y_{j\alpha/\beta}^n = y_j^n \exp\left(-\boldsymbol{ik}\cdot\boldsymbol{e}_{\alpha/\beta}\delta t\right), \end{cases} \tag{A5}$$

where $j = 1, 2, 3, 4$; $\boldsymbol{i}$ is the imaginary unit; and $\boldsymbol{k}$ denotes the vector of wavenumbers. As discussed in Sec. III, a set of linear equations can be obtained after differentiating Eqs. (A1)–(A4)

$$\overline{dy_j^*} = E_{jl}\cdot\overline{dy_l^n}, \quad j, l = 1, 2, 3, 4, \tag{A6}$$

where

$$(E_{jl})_{4\times4} = \left(\frac{\partial y_j^*}{\partial y_l}\right)^n = \begin{bmatrix} \partial_{11}^* & \partial_{12}^* & \partial_{13}^* & 0 \\ \partial_{21}^* & \partial_{22}^* & \partial_{23}^* & 0 \\ \partial_{31}^* & \partial_{32}^* & \partial_{33}^* & 0 \\ \partial_{41}^* & \partial_{42}^* & \partial_{43}^* & \partial_{44}^* \end{bmatrix} \tag{A7}$$

is the partial derivative matrix. Before we derive the specific partial derivatives, certain simplification can be applied to Eq. (A1)–(A4). Note that, for three thermal lattice velocity models considered in this paper, we can obtain the following relationships:

For D2Q4,

$$\begin{cases} \sum_{\beta=1}^{4} w_\beta^c \exp(-\boldsymbol{ik}\cdot\boldsymbol{e}_\beta\delta t) = \frac{1}{2}\cos(\theta_x) + \frac{1}{2}\cos(\theta_y), \\ \sum_{\beta=1}^{4} w_\beta^c e_{x\beta} \exp(-\boldsymbol{ik}\cdot\boldsymbol{e}_\beta\delta t) = -\frac{1}{2}\boldsymbol{i}\sin(\theta_x), \\ \sum_{\beta=1}^{4} w_\beta^c e_{y\beta} \exp(-\boldsymbol{ik}\cdot\boldsymbol{e}_\beta\delta t) = -\frac{1}{2}\boldsymbol{i}\sin(\theta_y). \end{cases} \tag{A8}$$

For D2Q5,

$$\begin{cases} \sum_{\beta=0}^{4} w_\beta^c \exp(-\boldsymbol{ik}\cdot\boldsymbol{e}_\beta\delta t) = \frac{1}{3} + \frac{1}{3}\cos(\theta_x) + \frac{1}{3}\cos(\theta), \\ \sum_{\beta=0}^{4} w_\beta^c e_{x\beta} \exp(-\boldsymbol{ik}\cdot\boldsymbol{e}_\beta\delta t) = -\frac{1}{3}\boldsymbol{i}\sin(\theta_x), \\ \sum_{\beta=0}^{4} w_\beta^c e_{y\beta} \exp(-\boldsymbol{ik}\cdot\boldsymbol{e}_\beta\delta t) = -\frac{1}{3}\boldsymbol{i}\sin(\theta_y). \end{cases} \tag{A9}$$

For D2Q9 (temperature field),

$$\begin{cases} \sum_{\beta=0}^{8} w_\beta^c \exp(-\boldsymbol{ik}\cdot\boldsymbol{e}_\beta\delta t) = \frac{4}{9} + \frac{2}{9}\cos(\theta_x) + \frac{2}{9}\cos(\theta_y) + \frac{1}{9}\cos(\theta_x)\cos(\theta_y), \\ \sum_{\beta=0}^{8} w_\beta^c e_{x\beta} \exp(-\boldsymbol{ik}\cdot\boldsymbol{e}_\beta\delta t) = -i\left[\frac{2}{9}\sin(\theta_x) + \frac{1}{9}\sin(\theta_x)\cos(\theta_y)\right], \\ \sum_{\beta=0}^{8} w_\beta^c e_{y\beta} \exp(-\boldsymbol{ik}\cdot\boldsymbol{e}_\beta\delta t) = -i\left[\frac{2}{9}\sin(\theta_y) + \frac{1}{9}\cos(\theta_x)\sin(\theta_y)\right]. \end{cases} \tag{A10}$$

For D2Q9 (flow field),

$$\begin{cases} \sum_{\alpha=0}^{8} w_\alpha^v \exp(-\boldsymbol{ik}\cdot\boldsymbol{e}_\alpha\delta t) = \frac{4}{9} + \frac{2}{9}\cos(\theta_x) + \frac{2}{9}\cos(\theta_y) + \frac{1}{9}\cos(\theta_x)\cos(\theta_y) \\ \sum_{\alpha=0}^{8} w_\alpha^v e_{x\alpha} \exp(-\boldsymbol{ik}\cdot\boldsymbol{e}_\alpha\delta t) = -i\left[\frac{2}{9}\sin(\theta_x) + \frac{1}{9}\sin(\theta_x)\cos(\theta_y)\right] \\ \sum_{\alpha=0}^{8} w_\alpha^v e_{y\alpha} \exp(-\boldsymbol{ik}\cdot\boldsymbol{e}_\alpha\delta t) = -i\left[\frac{2}{9}\sin(\theta_y) + \frac{1}{9}\cos(\theta_x)\sin(\theta_y)\right], \\ \sum_{\alpha=0}^{8} w_\alpha^v e_{x\alpha}^2 \exp(-\boldsymbol{ik}\cdot\boldsymbol{e}_\alpha\delta t) = \frac{2}{9}\cos(\theta_x) + \frac{1}{9}\cos(\theta_x)\cos(\theta_y), \\ \sum_{\alpha=0}^{8} w_\alpha^v e_{y\alpha}^2 \exp(-\boldsymbol{ik}\cdot\boldsymbol{e}_\alpha\delta t) = \frac{2}{9}\cos(\theta_y) + \frac{1}{9}\cos(\theta_x)\cos(\theta_y), \\ \sum_{\alpha=0}^{8} w_\alpha^v e_{x\alpha} e_{y\alpha} \exp(-\boldsymbol{ik}\cdot\boldsymbol{e}_\alpha\delta t) = -\frac{1}{9}\sin(\theta_x)\sin(\theta_y), \\ \sum_{\alpha=0}^{8} w_\alpha^v e_{x\alpha}^2 e_{y\alpha} \exp(-\boldsymbol{ik}\cdot\boldsymbol{e}_\alpha\delta t) = -\frac{1}{9}i\left[\cos(\theta_x)\sin(\theta_y)\right], \\ \sum_{\alpha=0}^{8} w_\alpha^v e_{x\alpha} e_{y\alpha}^2 \exp(-\boldsymbol{ik}\cdot\boldsymbol{e}_\alpha\delta t) = -\frac{1}{9}i\left[\sin(\theta_x)\cos(\theta_y)\right], \\ \sum_{\alpha=0}^{8} w_\alpha^v e_{x\alpha}^3 \exp(-\boldsymbol{ik}\cdot\boldsymbol{e}_\alpha\delta t) = -i\left[\frac{2}{9}\sin(\theta_x) + \frac{1}{9}\sin(\theta_x)\cos(\theta_y)\right], \\ \sum_{\alpha=0}^{8} w_\alpha^v e_{y\alpha}^3 \exp(-\boldsymbol{ik}\cdot\boldsymbol{e}_\alpha\delta t) = -i\left[\frac{2}{9}\sin(\theta_y) + \frac{1}{9}\cos(\theta_x)\sin(\theta_y)\right], \end{cases} \tag{A11}$$

where $\theta_x = k_x\delta t$ and $\theta_y = k_y\delta t$. By substituting Eq. (A11) into Eqs. (A1)–(A3), we can have the simplified formulations for the intermediate density and moments as follows:

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$$y_1^* = y_1^n\left[\frac{4}{9}+\frac{2}{9}\cos(\theta_x)+\frac{2}{9}\cos(\theta_y)+\frac{1}{9}\cos(\theta_x)\cos(\theta_y)\right] - iy_2^n\left[\frac{2}{3}\sin(\theta_x)+\frac{1}{3}\sin(\theta_x)\cos(\theta_y)\right] - iy_3^n\left[\frac{2}{3}\sin(\theta_y)+\frac{1}{3}\cos(\theta_x)\sin(\theta_y)\right] - \frac{y_2^n y_3^n}{y_1^n}\sin(\theta_x)\sin(\theta_y) + \frac{(y_2^n)^2}{y_1^n}\left[-\frac{2}{3}+\frac{1}{3}\cos(\theta_x)-\frac{1}{3}\cos(\theta_y)+\frac{1}{3}\cos(\theta_x)\cos(\theta_y)\right] + \frac{(y_3^n)^2}{y_1^n}\left[-\frac{2}{3}-\frac{1}{3}\cos(\theta_x)+\frac{1}{3}\cos(\theta_y)+\frac{1}{3}\cos(\theta_x)\cos(\theta_y)\right], \tag{A12}$$

$$y_2^* = -iy_1^n\left[\frac{2}{9}\sin(\theta_x)+\frac{1}{9}\sin(\theta_x)\cos(\theta_y)\right] + y_2^n\left[\frac{2}{3}\cos(\theta_x)+\frac{1}{3}\cos(\theta_x)\cos(\theta_y)\right] - y_3^n\left[\frac{1}{3}\sin(\theta_x)\sin(\theta_y)\right] - i\frac{(y_2^n)^2}{y_1^n}\left[\frac{2}{3}\sin(\theta_x)+\frac{1}{3}\sin(\theta_x)\cos(\theta_y)\right] - i\frac{(y_3^n)^2}{y_1^n}\left[-\frac{1}{3}\sin(\theta_x)+\frac{1}{3}\sin(\theta_x)\cos(\theta_y)\right] - i\frac{y_2^n y_3^n}{y_1^n}\left[\cos(\theta_x)\sin(\theta_y)\right], \tag{A13}$$

$$y_3^* = -iy_1^n\left[\frac{2}{9}\sin(\theta_y)+\frac{1}{9}\cos(\theta_x)\sin(\theta_y)\right] - y_2^n\left[\frac{1}{3}\sin(\theta_x)\sin(\theta_y)\right] + y_3^n\left[\frac{2}{3}\cos(\theta_y)+\frac{1}{3}\cos(\theta_x)\cos(\theta_y)\right] - i\frac{(y_2^n)^2}{y_1^n}\left[-\frac{1}{3}\sin(\theta_y)+\frac{1}{3}\cos(\theta_x)\sin(\theta_y)\right] - i\frac{(y_3^n)^2}{y_1^n}\left[\frac{2}{3}\sin(\theta_y)+\frac{1}{3}\cos(\theta_x)\sin(\theta_y)\right] - i\frac{y_2^n y_3^n}{y_1^n}\left[\sin(\theta_x)\cos(\theta_y)\right]. \tag{A14}$$

The formulations for the intermediate temperature vary with different thermal lattice velocity models. Substitution of Eqs. (A8)–(A11) into Eq. (A4) results in the following formulations:

For D2Q4 model,

$$y_4^* = y_4^n\left[\frac{1}{2}\cos(\theta_x)+\frac{1}{2}\cos(\theta_y)\right] - \frac{y_2^n y_4^n}{y_1^n}i\left[\frac{3}{2}\sin(\theta_x)\right] - \frac{y_3^n y_4^n}{y_1^n}i\left[\frac{3}{2}\sin(\theta_y)\right]. \tag{A15}$$

For D2Q5 model,

$$y_4^* = y_4^n\left[\frac{1}{3}+\frac{1}{3}\cos(\theta_x)+\frac{1}{3}\cos(\theta_y)\right] - \frac{y_2^n y_4^n}{y_1^n}\boldsymbol{i}\sin(\theta_x) - \frac{y_3^n y_4^n}{y_1^n}\boldsymbol{i}\sin(\theta_y). \tag{A16}$$

For D2Q9 model,

$$y_4^* = y_4^n\left[\frac{4}{9}+\frac{2}{9}\cos(\theta_x)+\frac{2}{9}\cos(\theta_y)+\frac{1}{9}\cos(\theta_x)\cos(\theta_y)\right] - \frac{y_2^n y_4^n}{y_1^n}i\left[\frac{2}{3}\sin(\theta_x)+\frac{1}{3}\sin(\theta_x)\cos(\theta_y)\right] - \frac{y_3^n y_4^n}{y_1^n}i\left[\frac{2}{3}\sin(\theta_y)+\frac{1}{3}\cos(\theta_x)\sin(\theta_y)\right]. \tag{A17}$$

With the above simplified expressions [Eqs. (A12)–(A17)] of the intermediate variables, we can obtain the formulations of all terms in matrix $\boldsymbol{E}$. Only the terms related to the intermediate temperature are illustrated below, expressions of other terms can be found in Ref. 16.

For D2Q4,

$$E_{41} = \left(\frac{\partial y_4^*}{\partial y_1}\right)^n = i\frac{y_2^n y_4^n}{(y_1^n)^2}\left[\frac{3}{2}\sin(\theta_x)\right] + i\frac{y_3^n y_4^n}{(y_1^n)^2}\left[\frac{3}{2}\sin(\theta_y)\right], \tag{A18}$$

$$E_{42} = \left(\frac{\partial y_4^*}{\partial y_2}\right)^n = -i\frac{y_4^n}{y_1^n}\left[\frac{3}{2}\sin(\theta_x)\right], \tag{A19}$$

$$E_{43} = \left(\frac{\partial y_4^*}{\partial y_3}\right)^n = -i\frac{y_4^n}{y_1^n}\left[\frac{3}{2}\sin(\theta_y)\right], \tag{A20}$$

$$E_{44} = \left(\frac{\partial y_4^*}{\partial y_4}\right)^n = \frac{1}{2}\cos(\theta_x)+\frac{1}{2}\cos(\theta_y) - i\frac{y_2^n}{y_1^n}\left[\frac{3}{2}\sin(\theta_x)\right] - i\frac{y_3^n}{y_1^n}\left[\frac{3}{2}\sin(\theta_y)\right]. \tag{A21}$$

For D2Q5,

$$E_{41} = \left(\frac{\partial y_4^*}{\partial y_1}\right)^n = i\frac{y_2^n y_4^n}{(y_1^n)^2}\sin(\theta_x) + i\frac{y_3^n y_4^n}{(y_1^n)^2}\sin(\theta_y), \tag{A22}$$

$$E_{42} = \left(\frac{\partial y_4^*}{\partial y_2}\right)^n = -i\frac{y_4^n}{y_1^n}\sin(\theta_x), \tag{A23}$$

$$E_{43} = \left(\frac{\partial y_4^*}{\partial y_3}\right)^n = -i\frac{y_4^n}{y_1^n}\sin(\theta_y), \tag{A24}$$

$$E_{44} = \left(\frac{\partial y_4^*}{\partial y_4}\right)^n = \frac{1}{3}+\frac{1}{3}\cos(\theta_x)+\frac{1}{3}\cos(\theta_y) - i\frac{y_2^n}{y_1^n}\sin(\theta_x) - i\frac{y_3^n}{y_1^n}\sin(\theta_y). \tag{A25}$$

For D2Q9,

$$E_{41} = \left(\frac{\partial y_4^*}{\partial y_1}\right)^n = i\frac{y_2^n y_4^n}{(y_1^n)^2}\left[\frac{2}{3}\sin(\theta_x)+\frac{1}{3}\sin(\theta_x)\cos(\theta_y)\right] + i\frac{y_3^n y_4^n}{(y_1^n)^2}\left[\frac{2}{3}\sin(\theta_y)+\frac{1}{3}\cos(\theta_x)\sin(\theta_y)\right], \tag{A26}$$

$$E_{42} = \left(\frac{\partial y_4^*}{\partial y_2}\right)^n = -i\frac{y_4^n}{y_1^n}\left[\frac{2}{3}\sin(\theta_x)+\frac{1}{3}\sin(\theta_x)\cos(\theta_y)\right], \tag{A27}$$

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$$E_{43}=\left(\frac{\partial y_4^*}{\partial y_3}\right)^n=-i\frac{y_4^n}{y_1^n}\left[\frac{2}{3}\sin(\theta_y)+\frac{1}{3}\cos(\theta_x)\sin(\theta_y)\right], \tag{A28}$$

$$\begin{aligned}E_{44}=\left(\frac{\partial y_4^*}{\partial y_4}\right)^n&=\frac{4}{9}+\frac{2}{9}\cos(\theta_x)+\frac{2}{9}\cos(\theta_y)+\frac{1}{9}\cos(\theta_x)\cos(\theta_y)\\&-i\frac{y_2^n}{y_1^n}\left[\frac{2}{3}\sin(\theta_x)+\frac{1}{3}\sin(\theta_x)\cos(\theta_y)\right]\\&-i\frac{y_3^n}{y_1^n}\left[\frac{2}{3}\sin(\theta_y)+\frac{1}{3}\cos(\theta_x)\sin(\theta_y)\right].\end{aligned} \tag{A29}$$

Similar procedure is considered in the corrector step. To avoid notational confusion, γ will be used later to denote the product of the gravitational acceleration g and the thermal expansion coefficient β, i.e., $\gamma=g\beta$. First, let us rewrite Eqs. (4)–(6) into the following forms:

$$\begin{aligned}y_1^{n+1}&=y_1^*+(\tau_v-1)(y_1^n-y_1^n),\\y_2^{n+1}&=y_2^*+(\tau_v-1)\left(\sum_\alpha e_{x\alpha}f_\alpha^{eq,*}-y_2^n\right),\\y_3^{n+1}&=y_3^*+(\tau_v-1)\left(\sum_\alpha e_{y\alpha}f_\alpha^{eq,*}-y_3^n\right)-\gamma y_1^n(y_4^n-T_0),\\y_4^{n+1}&=y_4^*+(\tau_c-1)\left(\sum_\beta g_\beta^{eq,*}-\mathrm{y}_4^n\right),\end{aligned} \tag{A30}$$

where $f_\alpha^{eq,*}$ and $g_\beta^{eq,*}$ are evaluated from the intermediate variables y_j^*, and similar simplified notations as Eq. (17) are used

$$\begin{cases}f_\alpha^{eq,*}=f_\alpha^{eq,*}(\boldsymbol{r}+\boldsymbol{e}_\alpha\delta t,\,t),\\g_\beta^{eq,*}=g_\beta^{eq,*}(\boldsymbol{r}+\boldsymbol{e}_\alpha\delta t,\,t),\\y_{j\alpha/\beta}^*=y_{j\alpha/\beta}^*(\boldsymbol{r}+\boldsymbol{e}_{j\alpha/\beta}\delta t,\,t).\end{cases} \tag{A31}$$

Following similar linearization procedure, i.e., differentiating Eq. (A30), we have the following relationships for corrector step:

$$\overline{d\boldsymbol{Y}^{n+1}}=\overline{d\boldsymbol{Y}^*}+\boldsymbol{M}\cdot\boldsymbol{S}\cdot\overline{d\boldsymbol{Y}^*}+H\cdot\overline{d\boldsymbol{Y}^n}, \tag{A32}$$

where $\boldsymbol{Y}$ denotes the variable vector and $\boldsymbol{M}$, $\boldsymbol{S}$, and $\boldsymbol{H}$ are certain matrices. Matrices $\boldsymbol{M}$ and $\boldsymbol{H}$ are explicitly written here

$$\boldsymbol{M}_{4\times4}=\begin{bmatrix}1&0&0&0\\0&\tau_v-1&0&0\\0&0&\tau_v-1&0\\0&0&0&\tau_c-1\end{bmatrix}, \tag{A33}$$

$$\boldsymbol{H}_{4\times4}=\begin{bmatrix}0&0&0&0\\0&-(\tau_v-1)&0&0\\-\gamma(y_4^n-T_0)&0&-(\tau_v-1)&-\gamma y_1^n\\0&0&0&-(\tau_c-1)\end{bmatrix}. \tag{A34}$$

Matrix $\boldsymbol{S}$ is defined as follows:

$$\boldsymbol{S}_{4\times4}=(S_{jl})_{4\times4}=\left(\frac{\partial S_j}{\partial y_l^*}\right), \tag{A35}$$

where

$$\begin{aligned}S_1&=y_1^n,\\S_2&=\sum_\alpha e_{x\alpha}f_\alpha^{eq,*},\\S_3&=\sum_\alpha e_{y\alpha}f_\alpha^{eq,*},\\S_4&=\sum_\beta g_\beta^{eq,*}.\end{aligned} \tag{A36}$$

For matrix $\boldsymbol{S}$, only the terms including the intermediate temperature are illustrated here, and other terms can be found in Ref. 16.

For D2Q4,

$$S_{41}=\left(\frac{\partial S_4}{\partial y_1^*}\right)=-i\frac{y_2^*y_4^*}{(y_1^*)^2}\left[\frac{3}{2}\sin(\theta_x)\right]-i\frac{y_3^*y_4^*}{(y_1^*)^2}\left[\frac{3}{2}\sin(\theta_y)\right], \tag{A37}$$

$$S_{42}=\left(\frac{\partial S_4}{\partial y_2^*}\right)=i\frac{y_4^*}{y_1^*}\left[\frac{3}{2}\sin(\theta_x)\right], \tag{A38}$$

$$S_{43}=\left(\frac{\partial S_4}{\partial y_3^*}\right)=i\frac{y_4^*}{y_1^*}\left[\frac{3}{2}\sin(\theta_y)\right], \tag{A39}$$

$$\begin{aligned}S_{44}=\left(\frac{\partial S_4}{\partial y_4^*}\right)&=\frac{1}{2}\cos(\theta_x)+\frac{1}{2}\cos(\theta_y)\\&+i\frac{y_2^*}{y_1^*}\left[\frac{3}{2}\sin(\theta_x)\right]+i\frac{y_3^*}{y_1^*}\left[\frac{3}{2}\sin(\theta_y)\right].\end{aligned} \tag{A40}$$

For D2Q5,

$$S_{41}=\left(\frac{\partial S_4}{\partial y_1^*}\right)=-i\frac{y_2^*y_4^*}{(y_1^*)^2}\sin(\theta_x)-i\frac{y_3^*y_4^*}{(y_1^*)^2}\sin(\theta_y), \tag{A41}$$

$$S_{42}=\left(\frac{\partial S_4}{\partial y_2^*}\right)=i\frac{y_4^*}{y_1^*}\sin(\theta_x), \tag{A42}$$

$$S_{43}=\left(\frac{\partial S_4}{\partial y_3^*}\right)=i\frac{y_4^*}{y_1^*}\sin(\theta_y), \tag{A43}$$

$$\begin{aligned}S_{44}=\left(\frac{\partial S_4}{\partial y_4^*}\right)&=\frac{1}{3}+\frac{1}{3}\cos(\theta_x)+\frac{1}{3}\cos(\theta_y)\\&+i\frac{y_2^*}{y_1^*}\sin(\theta_x)+i\frac{y_3^*}{y_1^*}\sin(\theta_y).\end{aligned} \tag{A44}$$

For D2Q9,

$$\begin{aligned}S_{41}=\left(\frac{\partial S_4}{\partial y_1^*}\right)&=-i\frac{y_2^*y_4^*}{(y_1^*)^2}\left[\frac{2}{3}\sin(\theta_x)+\frac{1}{3}\sin(\theta_x)\cos(\theta_y)\right]\\&-i\frac{y_3^*y_4^*}{(y_1^*)^2}\left[\frac{2}{3}\sin(\theta_y)+\frac{1}{3}\cos(\theta_x)\sin(\theta_y)\right],\end{aligned} \tag{A45}$$

$$S_{42}=\left(\frac{\partial S_4}{\partial y_2^*}\right)=i\frac{y_4^*}{y_1^*}\left[\frac{2}{3}\sin(\theta_x)+\frac{1}{3}\sin(\theta_x)\cos(\theta_y)\right], \tag{A46}$$

$$S_{43}=\left(\frac{\partial S_4}{\partial y_3^*}\right)=i\frac{y_4^*}{y_1^*}\left[\frac{2}{3}\sin(\theta_y)+\frac{1}{3}\cos(\theta_x)\sin(\theta_y)\right], \tag{A47}$$

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$$S_{44} = \left(\frac{\partial S_4}{\partial y_4^*}\right) = \frac{4}{9} + \frac{2}{9}\cos(\theta_x) + \frac{2}{9}\cos(\theta_y) + \frac{1}{9}\cos(\theta_x)\cos(\theta_y) - i\frac{y_2^*}{y_1^*}\left[\frac{2}{3}\sin(\theta_x) + \frac{1}{3}\sin(\theta_x)\cos(\theta_y)\right] - i\frac{y_3^*}{y_1^*}\left[\frac{2}{3}\sin(\theta_y) + \frac{1}{3}\cos(\theta_x)\sin(\theta_y)\right]. \quad \text{(A48)}$$

Together with the matrix $\boldsymbol{E}$ in Eq. (A6) and the formulations of y_j^* in Eqs. (A12)–(A17), we can derive

$$\overline{d\boldsymbol{Y}^{n+1}} = [(\boldsymbol{I} + \boldsymbol{M}\cdot\boldsymbol{S})\cdot\boldsymbol{E} + \boldsymbol{H}]\cdot\overline{d\boldsymbol{Y}^{n}} = \boldsymbol{G}\cdot\overline{d\boldsymbol{Y}^{n}}, \quad \text{(A49)}$$

where $\boldsymbol{G}$ is the overall characteristic matrix for the stability analysis of the STLBM. Note that the expressions of the elements of $\boldsymbol{G}$ vary with the thermal lattice velocity model employed.

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